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Revision 7

Signa EXCITE Block Diagrams
& Supplemental Schematics

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Technical Publications

Direction 2333494

Revision 8

Signa[®] EXCITE Block Diagrams & Supplemental Schematics

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GE Medical Systems

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America)

GE Medical Systems — Europe: Telex 261794
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- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
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- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
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REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
A. . . .	Ap 19, 2002	Preliminary release.
0. . . .	Jul 19, 2002	Initial release.
1. . .	Dec 12, 2002	Changes and corrections to match production.
2. . .	Feb 12, 2003	Changes and corrections to match production and including CQA corrections.
3. . .	Feb 26, 2003	Changes and corrections to match production. Specifically; Added SMC and its definition to Overview TAB page 3–5. Changes to Operator Workspace TAB page 1–1. Changes to MGD/RRF/RF (1.5T) TAB pages 1–1 & 1–2.
4. . . .	Sep 4, 2003	Changed name of manual and headers on all pages from Signa Release 10.x to Signa EXCITE. Added Linux information to System and Operator Workspace Tabs. Revised Operator Workspace tab with SCIM information.
5	Nov 7, 2003	System Tab – Updated to SRI–2 and 8 Channel Switch for Linux section. Operator Workspace Tab – Added connection information to System Cabinet J26. Patient Handling Tab – Added Linux section. MGD/RRF/RF Tab – Updated entire section with Linux information. Cooling System Tab – Added MRCC and GWHX cooling system information.
6 . . .	Apr 19, 2004	Operator Workspace Tab – Revised ethernet connection to Host computer by indicating left and right port connections. MGD/RRF/RF Tab – Revised quad cable to 1/4λ and revised note on sheet 8, Revised motherboard signal informatino on sheet 9A.
7 . . .	May 31, 2005	System Tab – Removed Octane SRF and SRFD2 cabinet info (refer to EXCITE 2 in section 2 for info). Added EXCITE 2 to section 2. Patient Handling Tab – Added EXCITE 2 to titles in Section 2. MGD/RRF/RF Tab – Added PIM board to MGD Chassis. Added EXCITE 2 to several titles.
8	Dec 17, 2008	PATIENT HANDLING Tab – Revised E–Stop circuit voltage from 24V to 12V in Teal PDU (PQR 13207104).

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* This revision number/letter corresponds to the indicated document's revision control system.

CONTENTS

OVERVIEW

Purpose
Standard Conventions
Abbreviations



SYSTEM

Overall System – Octane
Overall System – Linux



OPERATOR WORKSPACE

Operator Workspace – Octane
Operator Workspace – Linux



PATIENT HANDLING

Patient Handling – Octane
Patient Handling Block – Linux



GRADIENT

Gradient Block Diagram (ACGD)



PDU

PDU Block Diagrams

Note:

Phoenix PDU found in Vendor Manuals only. (SEE Service Methods CDRM)



MGD/RRF/RF (1.5T)

MGD/RRF/RF Block Diagrams (1.5T)
RF Supplemental Schematics (1.5T)



COOLING SYSTEM

Water Chillers Block Diagrams

Note:

For all Chillers and Cooling System variations (SEE Vendor Manuals on MR Service Methods CDRM)



OVERVIEW

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SECTION 1 – PURPOSE

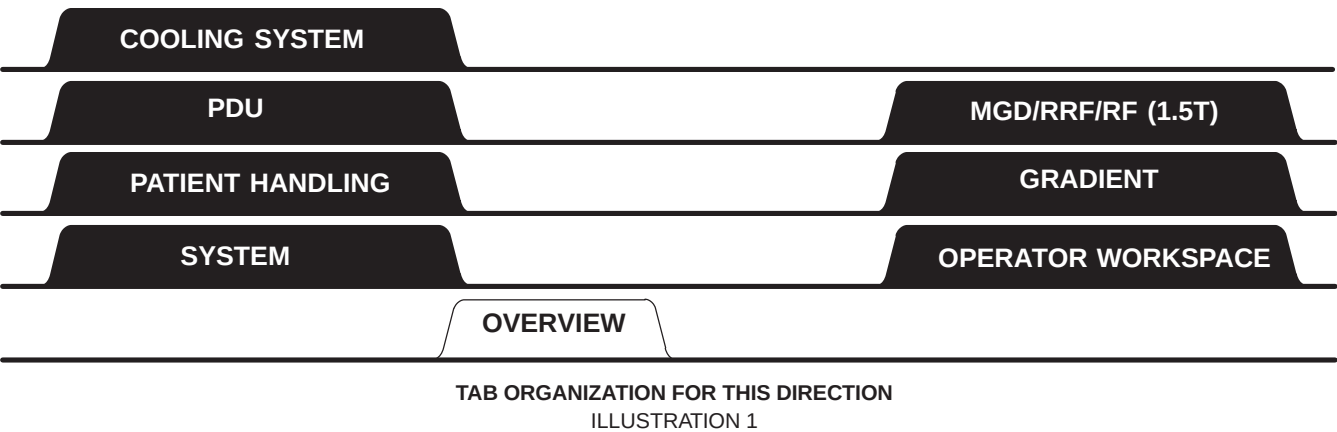
Note

This revision of Direction 2333494 is compatible with EXCITE hardware.

The purpose of this document is to provide an overview of the hardware which makes up the MR Signa System, and a quick means to troubleshoot major signals within the system.

It is recommended that you page through this manual to become familiar with its content and organization, beginning with System Tab, OVERALL SYSTEM. This section shows how the major subsystems are grouped by cabinet, and the main communication lines between them.

The remaining block diagrams are organized into sections which correspond to major subsystems. Some diagrams show hardware from another subsystem in order to complete an entire signal flow route. For such exceptions, the board/module title of the circuitry shown will generally help identify the applicable subsystem. See Illustration 1–1 for tab organization of this manual.



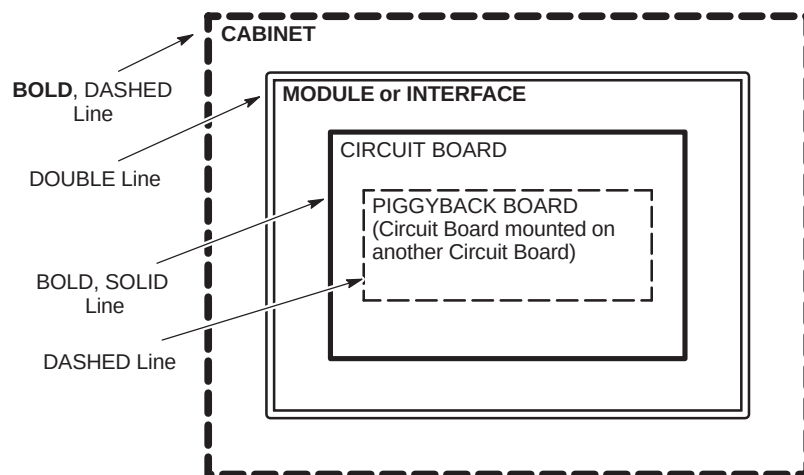
These block diagrams, along with provided diagnostics, are intended to be the primary tools for troubleshooting the MR System from a system or subsystem level on down to the specific module or board level. Since some of the RF subsystem block diagrams are presently insufficient for troubleshooting to the FRU (Field Replaceable Unit) level, schematics have been provided to supplement those block diagrams. When using the diagnostics, the block diagrams can provide a better understanding of what circuitry the diagnostics are testing, and how to further isolate a problem if required.

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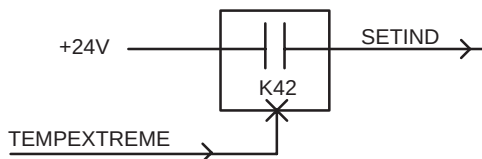
SECTION 2 – STANDARD CONVENTIONS

The following standard conventions are used within the block diagrams:

- The type of line used to enclose a major box on a block diagram indicates the following:



- The component designator for cabinets, modules, circuit boards, etc. is enclosed in "()" after or below their name on the block diagram.
- Logic levels: A "0" indicates a logic "low" level (typ. <0.8VDC), a "1" indicates a logic "high" level (typ. > 2.0VDC).
- An "*" is used to identify an "active low" signal (e.g. "INT*" signal when at a logic "0" indicates that an interrupt is present; a logic "1" level indicates that no interrupt is present).
- A "#/" indicates the number of wires in a cable (e.g. $\frac{9}{\text{---}}$ indicates 9 wires).
- Signal flow is generally from left to right on a block diagram. Arrows are used to identify exceptions to this rule and to aid in identifying input/output signal lines on the diagram.
- The "x" symbol (e.g. $\text{---}x$) shown on boxes in a block diagram are used to indicate that the function within the box is enabled by the "active" signal on this input line.



For this example, when signal TEMPEXTREME goes active ("1"), relay K42 will close (become enabled) and the signal SETIND now becomes active to set a temperature indicator.

SECTION 3 – ABBREVIATIONS

Note

This list has been compiled from a wide variety of sources. Not all of these abbreviations will apply to your system. For an electronic copy of over 1000 acronyms, see *Direction 2124201–2*.

A

AC = Alternating Current
 ACGD = Advanced Concept Gradient Driver
 ACK = Acknowledge
 ACQ = Acquisition
 A/D = Analog-to-Digital
 ADC = Analog to Digital Converter
 ADD or ADDR = Address
 ADJ = Adjust
 ADV = Advance
 AGC = Automatic Gain Control
 AGP = Applications Gateway Processor (Board)
 AI = Applications Interface, Artificial Intelligence
 ALGN = Align
 ALU = Arithmetic Logic Unit
 AM = Amplitude Modulation
 AMP = ampere, amplifier
 A/N = Alpha-numeric
 ANSI = American National Standards Institute
 AP = Array Processor
 APM = Analog Power Monitor
 APS = Auto Pre-Scan or Acquisition Processing System (Board)
 APSGT = Auto Pre-Scan Gain Control
 ARCH = Archive/Remove Process
 ASC = Automated Service Center
 ASCII = American Standard Code for Information Interchange
 ASM or ASSY = Assembly
 ASM = Analog Service Module
 ATTEN = Attenuator
 AUTO = Automatic
 AUX = Auxiliary

B

B = magnetic field
 BATT = Battery
 BB = Buffered Data or Broadband
 BC = Body Coil

BD = Board
 BDS = Buffered Data Strobe
 BLK = Black
 BNC = BNC Connector
 BRM = Body Resonance Module
 BRT = Brightness
 BUF = Buffer
 B/W = BandWidth

C

C = centigrade, Celsius
 CAB = Cabinet
 CAL = Calibration
 CALC = Calculate
 CAN = Control Area Network
 CATTEN = Course Attenuation
 CB = Circuit Breaker
 CCW = Counter Clockwise
 CD-ROM = Compact Disk Read-Only Memory
 CE = Chip Enable
 CF = Center Frequency
 CFA = Center Frequency Adjust
 CHAN or CHNL = Channel
 CHK = Check
 CKT = Circuit
 CLR = Clear
 CMD = Command
 CMMR = Common Mode Rejection Ratio
 CMOS = Complimentary Symmetry Metal Oxide Semiconductor
 CMP = Compare or Compressor
 CNT = Contrast
 COAX = Coaxial
 COF = Cut Off Frequency
 CONFIG = Configuration
 CONT = Continuous
 CONV = Converter
 CPD = Communication Pin Driver
 CPU = Central Processing Unit

CRT = Cathode-Ray Tube
 CTL = Clock Time Control
 CTRL = Control
 CV = Configuration Variable
 CW = Clockwise
 CYL = Cylinder

D

D/A = Digital-to-Analog
 DAC = Digital to Analog Converter
 dB = Decibel
 dBm = Decibels above or below one Milliwatt
 DC = Direct Current
 DCB = Driver Control Board
 DCKSTP = Dock Stop
 DD = Dynamic Disable
 DDS = Direct Digital Synthesizer
 DDSB = Dynamic Disable Switch Box
 DEC = Decimal
 DEG = Degree(s)
 DEMUX = Demultiplexer
 DEV = device
 DIA = Diagnostic Imaging Accessories
 DIAG = Diagnostic
 DIFF = Differential
 DIG = Digit
 Dig B0 = Digital B0 Adjustment
 DIP = Dual In-line Package
 DIR = Direction
 DIS = Disable
 DMA = Direct Memory Access
 DMAC = Direct Memory Access Controller
 DMM = Digital Multi-Meter
 DPR or DPRAM = Dual Port Random Access Memory
 DQA = Daily Quality Assurance
 DRAM = Dynamic Random Access Memory
 DRC = Dynamic Ram Controller
 DRF = Digital Receive Filter (Board)
 DRVR = Driver
 DSP = Digital Signal Processor
 DVM = Digital Volt Meter
 DWG = Drawing

E

ECC = Extended Cursor Control or Error Checking and Correction
 ECG = Electro-Cardiogram

EEPROM = Electrically Erasable Programmable Read Only Memory
 EFB = Envelop Feedback
 EPI = Echo Planer Imaging
 e.g. = for example
 EM or EMERG = Emergency
 EMI = Electro-Magnetic Interference
 EN or ENA = Enable
 ENV = Envelope
 EOS = End Of Sequence
 EOT = End Of Travel
 EOW = End Of Waveform
 EPO = Emergency Power Off
 EPROM = Erasable Programmable Read Only Memory
 ERU = Emergency Rundown Unit
 EXER = Exerciser

F

f = farad
 F = Fahrenheit, Fuse (designator)
 FBD = Functional Block Diagram
 FBK = Feedback
 FC = footcandle
 Fc = Center Frequency
 FDBK = Feedback
 FDO = Future Delivery Order
 FE = Field Engineer
 FET = Field-Effect Transistor
 FF or FIFO = First In – First Out
 FFD = Full Field Distortion
 FFT = Fast Fourier Transform
 FID = Free Induction Decay
 FIDAT = FIFO In Data
 FIFO or FF = First In – First Out
 FILT = Filter
 FM = Frequency Modulated
 FMI = Field Modification Instruction
 FOV = Field Of View
 FREQ = Frequency
 FRNT = Front
 FRU = Field Replaceable Unit
 ft = foot
 FT = Feet
 FTP = File Transfer Protocol

G

G = gauss
 G/A = Gradient Amplifier
 GEN = Generator
 GND = Ground
 GRAD = Gradient
 GRN = Green
 GRY = Grey
 GP = Gradient Processor

H

H = henry
 H = Hexadecimal
 HC = Head Coil
 HD = Head
 He = Helium
 He-ALARM = Helium Alarm
 HEX = Hexadecimal
 He-WARNING = Helium Warning
 HORIZ or HORZ = Horizontal
 HPC = Hardware and Protocol Control
 HPFS = Host Platform Software
 hr = hour
 HSN = High Speed Network
 HSS (D) = High Speed Serial Data Link
 HTR = Heater
 HV = High Voltage
 HV PSU = High Voltage Power Supply Unit
 HVAC = Heating, Ventilation, & Air Conditioning
 Hz = hertz

I

I = Current or Phase or Real
 IAMP = Amplifier Current (Gradient)
 ICD = Installation Certification Document
 ICL = Coil Overcurrent (Gradient)
 ID = inner diameter
 ID = Identity, Identifier
 i.e. = that is to say
 IEC = International Electrical Code
 IEEE = Institute of Electrical & Electronics Engineers
 I/F = Interface
 in. = inch
 INFO = Information
 INS, INST, or INSTR = Instruction
 INT or INTR = Interrupt
 I/O = Input/Output

IP = Image Processor or Plate Current (RF amp) or
 Instruction Processor or Internet Protocol
 IPA = Intermediate Power Amplifier
 IPC = Interprocess Communication
 IQ = Image Quality
 IRF = Interface and Remote Functions (Board)
 IRF I/O = Interface and Remote Functions
 Input/Output (Board)
 IRQ = Interrupt Request
 ISA = Interrupt Start Address
 ISI = Inter-Sequence Interrupt
 ISO = Isolator or Isolation
 IT-MGD = Image Transfer-MDG (Board)
 IT-Host = Image Transfer-Host (Board)
 IVI = Interactive Vascular Imaging

J

J = Jack Connector (designator)
 JP = Jumper

K

K = Kelvin (unit of temperature) or Kilo (Thousand)
 KB = one thousand bytes of memory
 KCal = kiloCalorie
 kg = kilogram
 kHz = kilohertz
 KVA = kilo Volt Amperes
 kW = Kilowatt

L

L or LFT = Left
 LAN = Local Area Network
 lb = pound
 LBK = Loopback
 LCA = Logic Cell Arrays
 LED = Light Emitting Diode
 LFC = Load From Cold
 LHe = Liquid Helium
 LIN = Linear
 LM or LMK = Landmark
 LMP = Lamp
 LNK = Link
 LPCA = Low Profile Carriage Assembly
 LO = Local Oscillator
 LOEP = List of Effective Pages
 LONG = Longitudinal
 LVLE = Low Voltage, Low Energy

M

m = meter
 M = mega – one million
 mA = millamps
 MAG = Magnet
 MB = megabyte
 MC = Multi-Coil
 MCD = Multi-Coil switch Driver
 MDS = Multidrop Serial interface
 MEM = Memory
 Mfg. = Manufacturing
 MFU = Multi-Format Unit
 MGD = Multi-Generational Data acquisition (Chassis)
 MGMNT = Management
 MHz = megahertz
 mm = millimeter
 MOD = Modifier
 MON = Monitor
 MOSFET = Metal Oxide Semiconductor Field Effect Transistor
 MOV = Metal Oxide Varistor
 MPS = Manual Pre-Scan
 MR = Magnetic Resonance
 MSG = Message
 MTR = Motor
 MUX = Multiplexer
 mV = millivolts
 MW = megawatt
 mW = milliwatt

N

N = Negative
 N2 = Nitrogen
 n.c. = normally closed
 NEC = National Electrical Code
 Ni-Cad = Nickel Cadmium
 n.o. = normally open
 NO. = Number
 NOPROC = No Processing
 NOREC = No Reconstruction
 nS = Nanosecond

O

O2 = oxygen
 OC = Operator's Console
 OD = outside diameter
 OM = Oxygen Monitor
 OPI = Oblique Plane Imaging

OPTO = Optical
 ORG = Orange
 OT = Overtemp
 OV = Over voltage
 OVR = Over
 OVRLD or OL = Overload
 oz = ounce
 OW = Operator Workspace

P

P = Positive
 PA = Patient or Power Amplifier
 PAC = Physiological Acquisition Controller
 PAL = Programmable Array Logic or Patient Alignment Lights
 PAN = Primary Archive Node
 PAT = Patient
 PCI = Programmable Communications Interface or Peripheral Component Interconnect
 PCT = Pulse Controller Task
 PDI = Product Delivery Instruction
 PDU = Power Distribution Unit
 PEN = Penetration
 PIN = Positive-Intrinsic-Negative
 PIO = Programmed Input/Output
 PLL = Phase Lock Loop
 PIXEL = Picture Element
 PM = Phase Modulation, Periodic Maintenance
 PMC = PCI Mezzanine Connector
 P/N = Positive/Negative
 PNL = Panel
 POS = Position
 POT = Potentiometer
 PP = Penetration Panel
 PPC = Power PC
 P-P = Peak to Peak
 ppm = parts per million
 PREAMP = Preamplifier
 PROC = Process, Processor
 PROG = Program, -ed, -able
 PROM = Programmable Read Only Memory or Program
 PS = Power Supply
 PSD = Pulse Sequence Database
 PSG = Pulse Sequence Generator
 PSU = Power Supply Unit
 PSS = Power Supply System Board
 PT = Patient Transport or Patient Table

PUR = Purple
 PVC = Polyvinylchloride
 PW = Pulse Width
 PWA = Printed Wiring Assembly
 PWB = printed wiring board
 PWR = Power

Q

Q = Quadrature, Quality Factor or Imaginary
 Qty = Quantity
 QUAD = Quadrature or Quadrant

R

R or RT = Right
 RAM = Random Access Memory
 RCV = Receive
 RD = Read
 RDY = Ready
 RECON = Reconstruction
 REF = Reference or Reflected
 REG = Register or Regulator
 RET or RTN = Return
 REV = Revision or Reverse
 RF = Radio Frequency
 RFAMP = Radio Frequency Amplifier
 RFA = RF Amplifier
 RFI = Radio Frequency Interference or RF Interface
 (Module)
 RGB = Red Green Blue
 RLA or RLY = Relay
 rms = root mean square
 ROI = Region of Interest
 ROM = Read Only Memory
 RPC = Remote Control Panel
 RRF = Remote RF (Chassis)
 RSR = Register
 RST = Reset
 R/W = Read/Write
 RX = Receive
 RXD = Receive Data

S

S = Switch (designator)
 SACK = Slave Acknowledge
 SAG = Sagittal
 SAR = Specific Absorption Rate
 S/C = Superconducting
 SCA = Scan Control Assembly
 SCP = Scan Control Processor (Board)

SCR = Silicon Controlled Rectifier
 SCSI = Small Computer System Interface
 SEC = second
 SEG = Segment
 SEL = Select
 SEPI = Spin Echo Planer Imaging
 SGA = Switchable Gradient Amplifier
 SGI = Silicon Graphics Inc.
 SIG = Signal
 SII = "S-2" (type of magnet)
 SIII = "S-3" (type of magnet)
 SIMMS = Single In-line Memory Module
 SIP = Single In-line Package
 SL = Slice
 SMC = Status Monitoring and Control
 SMPTE = Society of Motion Picture and Television
 Engineers
 SN = Service Note
 SNR = Signal-to-Noise Ratio
 SPI = Serial Peripheral Interface
 SPT = System Performance Test
 SPU = Signal Processing Unit
 SRAM = Static Random Access Memory
 SRF = Solid State RF Amplifier SGA or Scalable RF
 (Cabinet)
 SRF/TRF = Sequencer Related Function / Trigger and
 Rotation Function (Board)
 SRI = Scan Room Interface
 SRV = Service
 SSM = System Support Module
 SSP = Sequence Synchronous Protocol
 SST = Small Sample Test
 STAB = Stability
 STAT = Status
 STD = Standard
 STD DEV = Standard Deviation
 STIF Bd = SRF/TRF InterFace Board
 SUB-D = Subminiature D (connector)
 SW = Switch or Software
 S/W = Software
 SYNC = Synchronous
 SYS = System
 SYSID = System Identification

T

T = tesla
TAC = Twin Accessory Cabinet
TBD = To Be Determined
TBL = Table
TEMP = Temperature
TFU = Thyristor Firing Unit
TG = Transmit Gain
TGWC = TwinSpeed Gradient Water Cooler
TNS = Transient Noise Suppressor
TP = Test Point (designator)
TR or T/R = Transmit–Receive
TRMAP = Transmit/Receive Map
TRM = Twin Coil Resonator Module
TS = Terminal Strip
TSCC = TwinSpeed Cooling Cabinet
TST = Test
TTL = Transistor to Transistor Logic
TX = Transmit
TXD = Transmit Data

U

UART = Universal Asynchronous Receiver–Transmitter
uf = microfarad
UFI = Ultra Fast Imaging
UNBLK = Unblank
uP = Microprocessor
UPLMT = Up Limit
USART = Universal Synchronous/Asynchronous
Receiver–Transmitter
uS or usec = microsecond
UTNS = Universal Transient Noise Suppressor (Board)
UV = Under voltage

V

VAC = Volts Alternating Current
VAR = Variable
VBUS = Bus Voltage
VCR = Video Cassette Recorder
VCNTRL = Control Voltage for 8645 Techrons
VDC = Volts Direct Current
VERT = Vertical
Vp–p = Volts Peak–to–Peak
VRMS = Volts, Root Mean Square
VTR = Video Tape Recorder

W

WARP = Waveform And Rotation Processor
WC = Water Chiller
WR = Write
WHT = White
WIM = Workspace Interface Module
WND = Window
WORM = Write Once Read Many
WRT = Write

X

XCVR = Transceiver
XFMR = Transformer
XMT or XMIT = Transmit

Y

YEL = Yellow

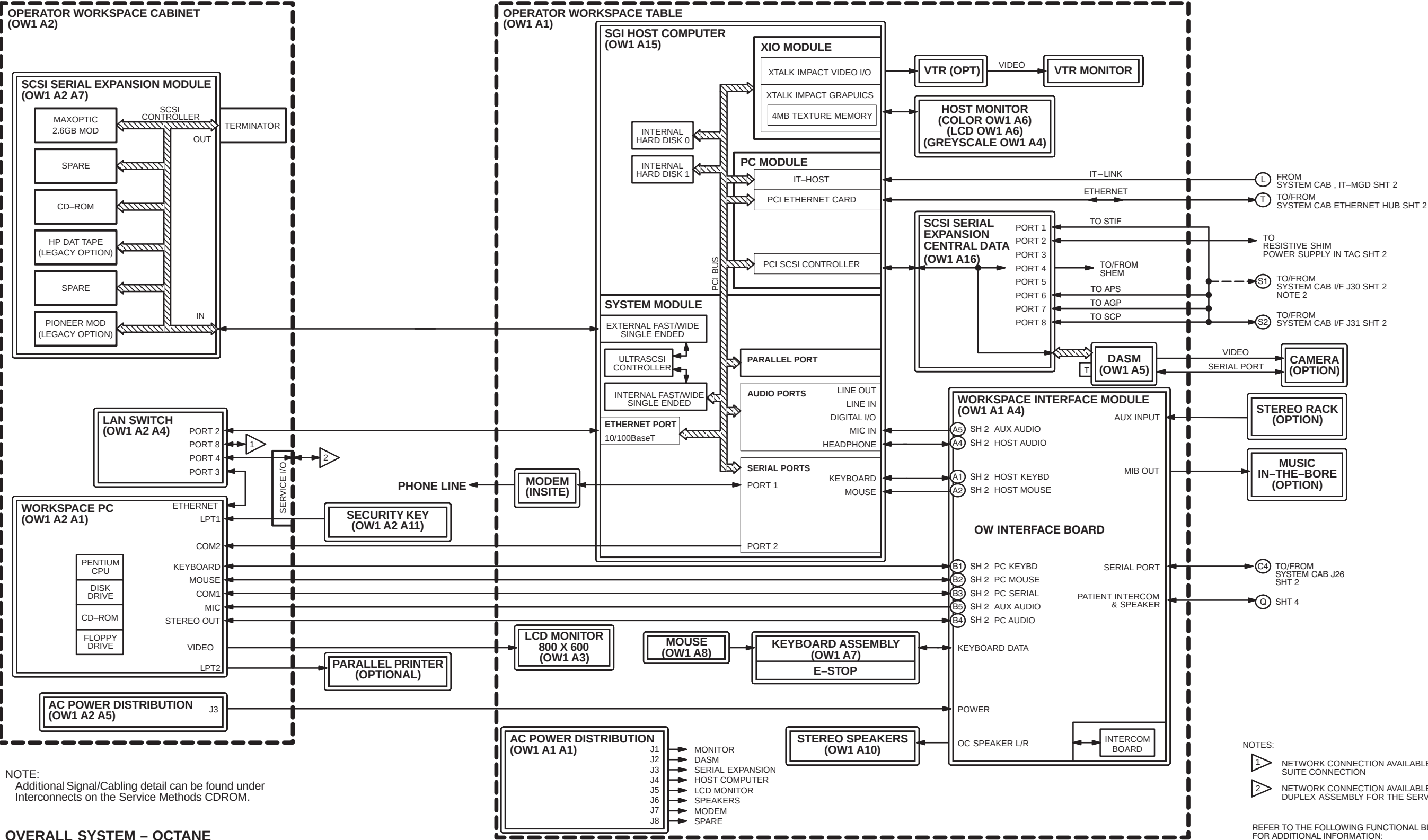
SYSTEM

TABLE OF CONTENTS

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1 – OVERALL SYSTEM – OCTANE	1-1
2 – OVERALL SYSTEM – LINUX	2-1

NOTE:
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

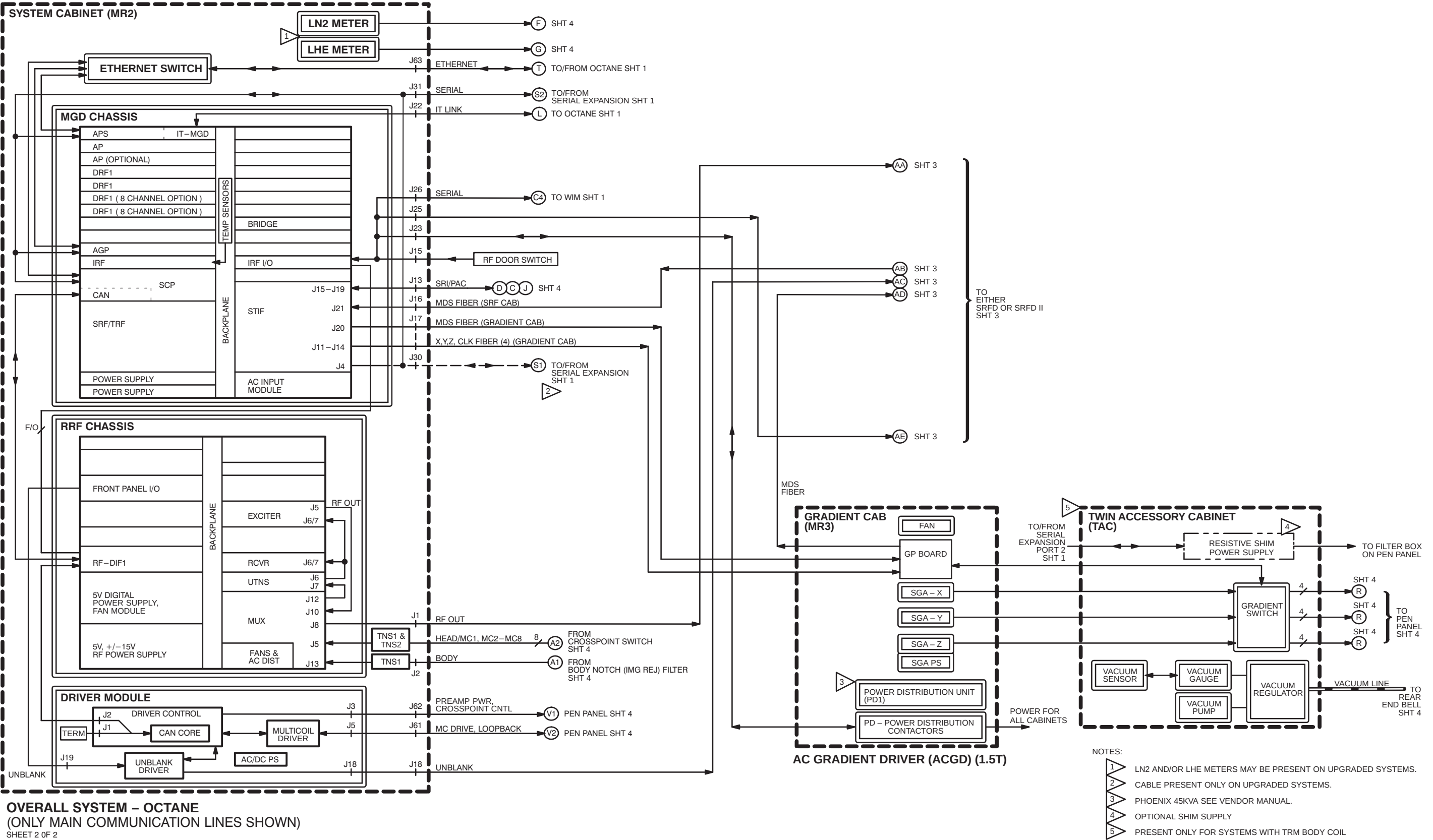
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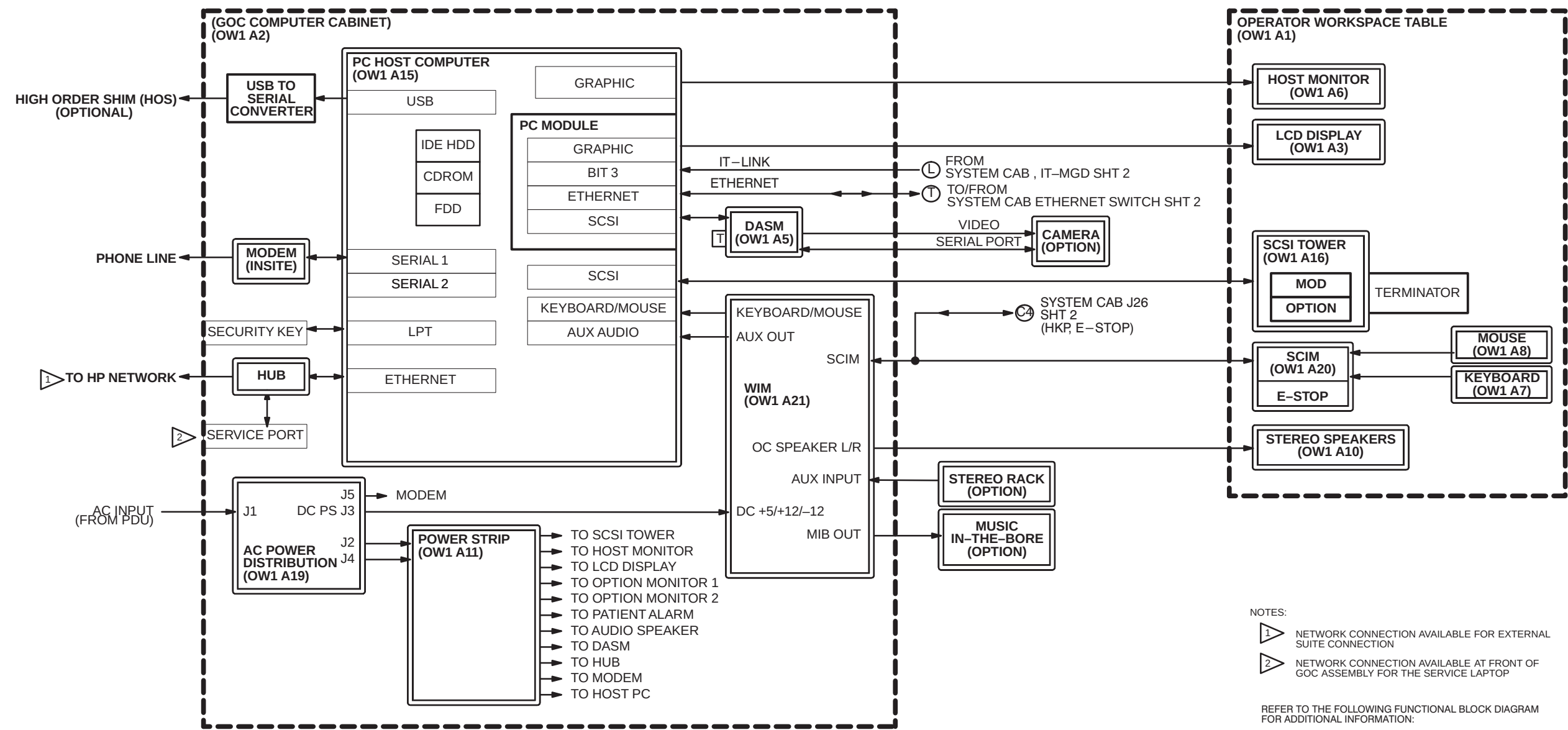


OVERALL SYSTEM – OCTANE
(OPERATOR WORKSPACE CABINET/TABLE – OCTANE)

SHEET 1 OF 2

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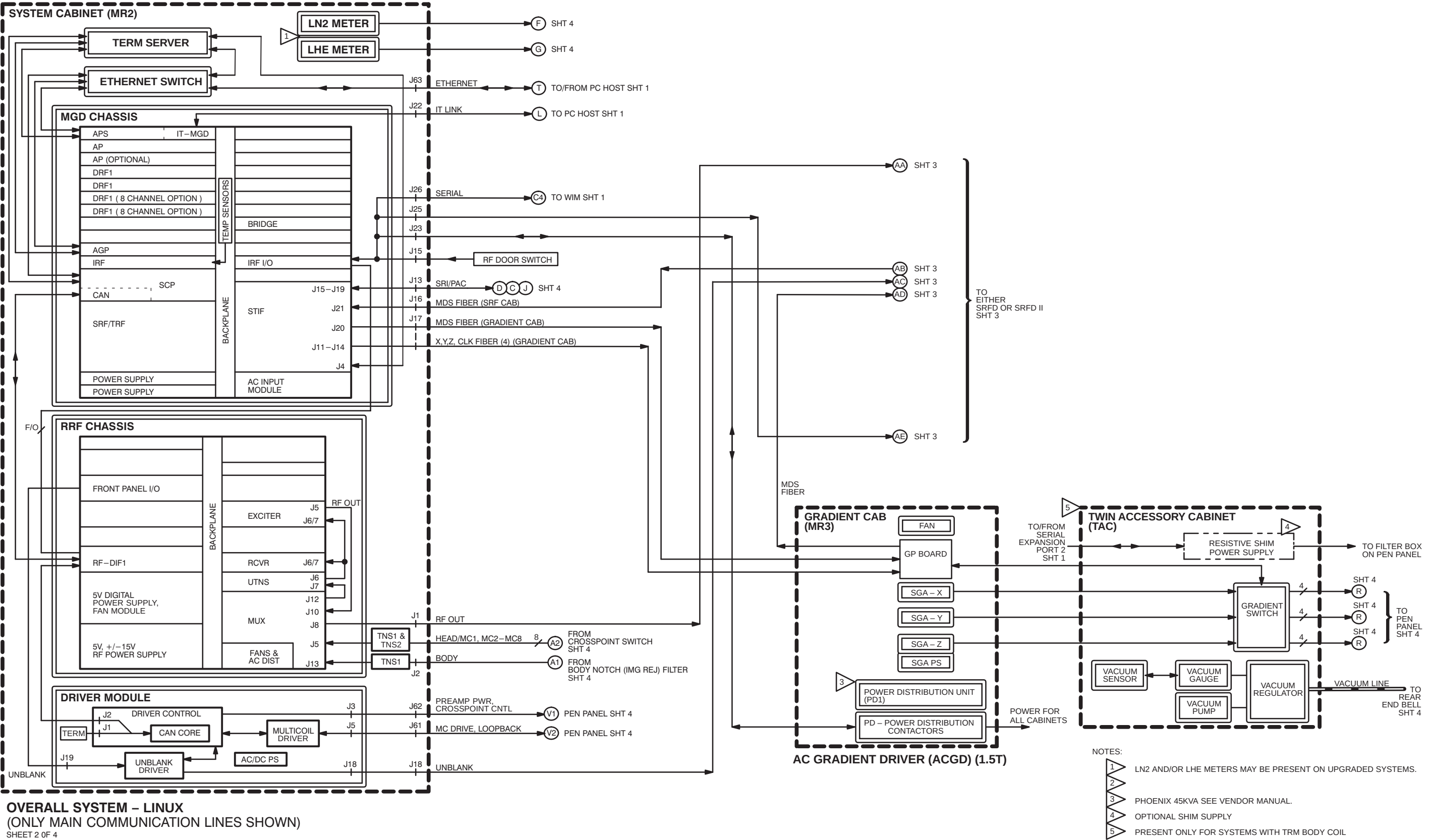


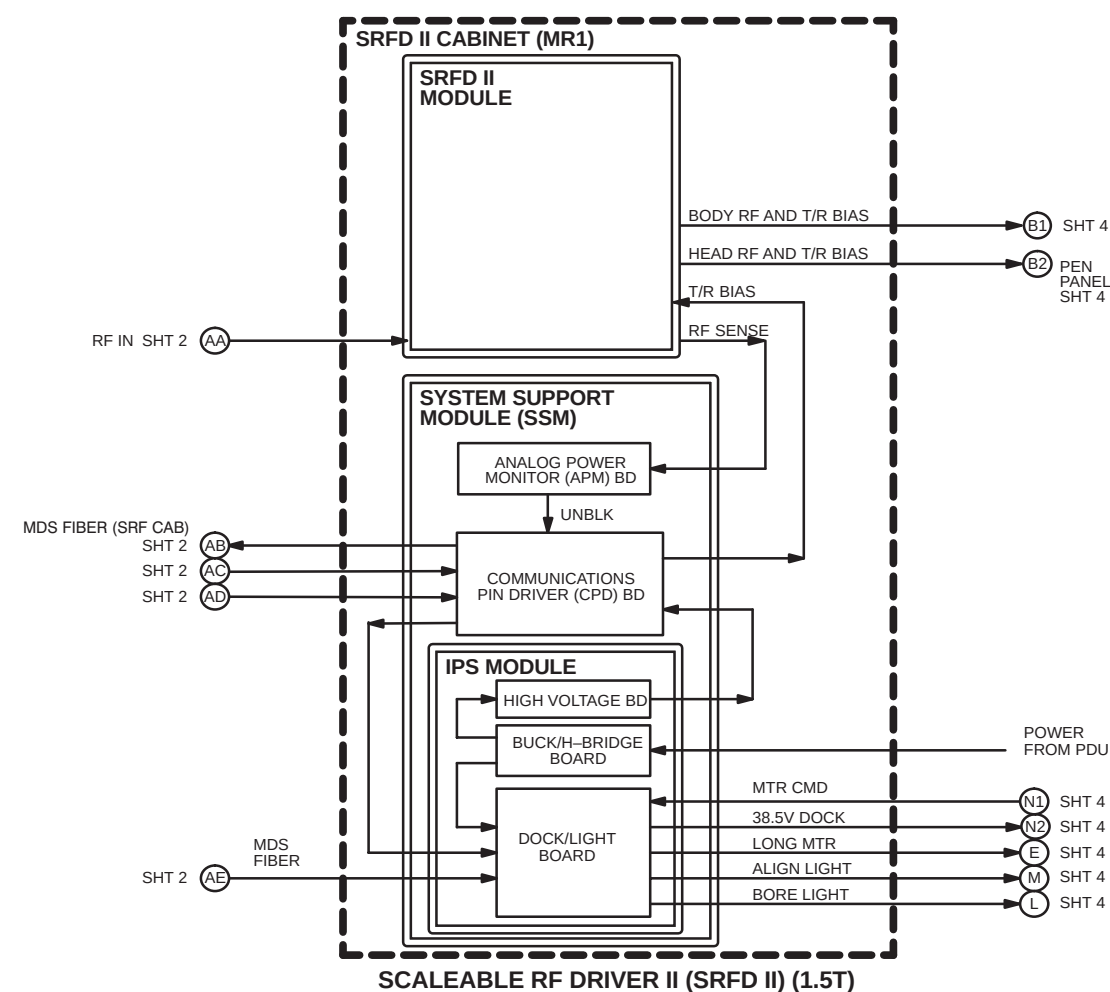


NOTE:
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

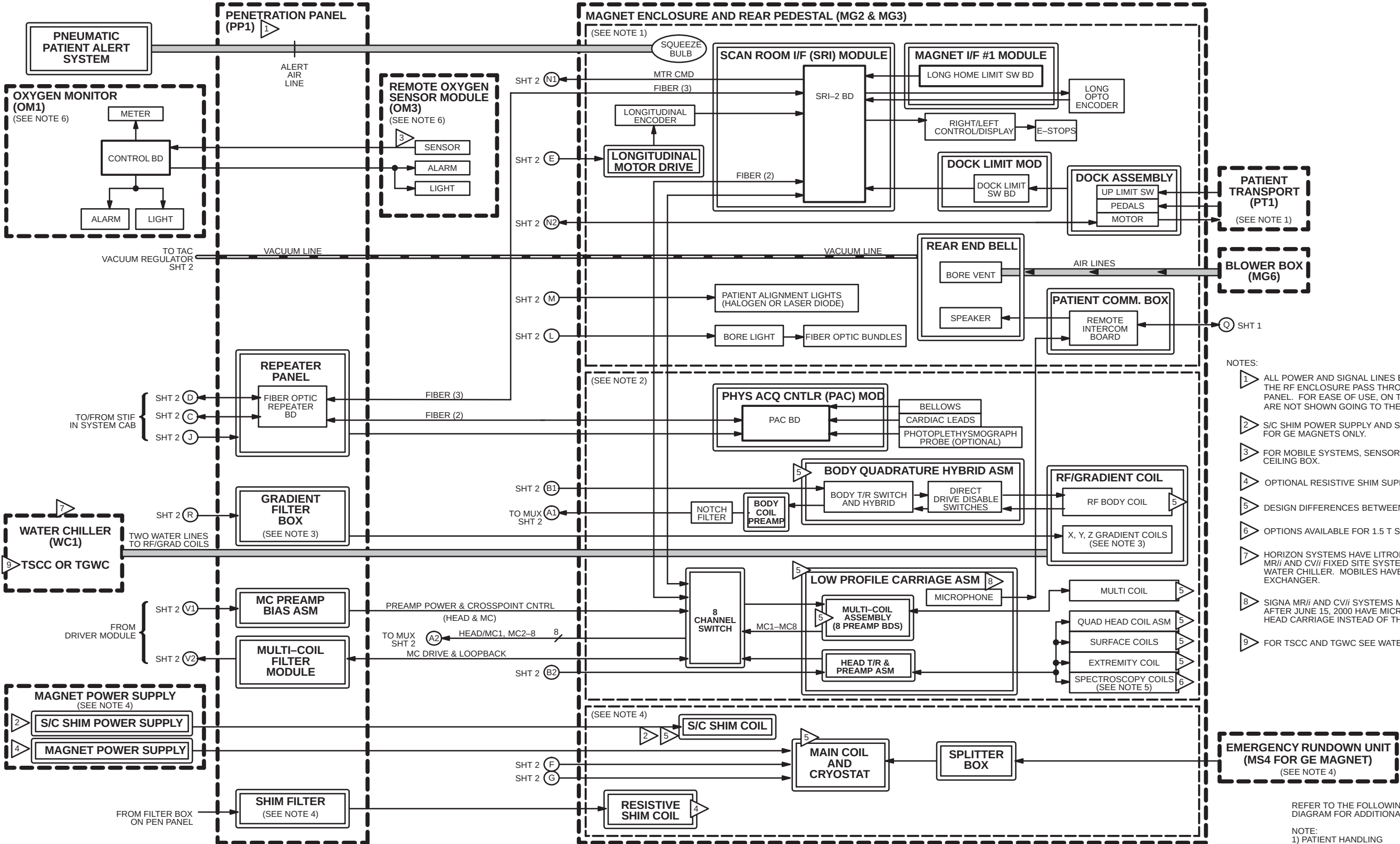
OVERALL SYSTEM – LINUX
(OPERATOR WORKSPACE CABINET/TABLE – LINUX)

SHEET 1 OF 4





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- NOTES:
- ALL POWER AND SIGNAL LINES ENTERING AND LEAVING THE RF ENCLOSURE PASS THROUGH THE PENETRATION PANEL. FOR EASE OF USE, ON THIS DIAGRAM, THEY ARE NOT SHOWN GOING TO THE PENETRATION PANEL.
 - S/C SHIM POWER SUPPLY AND S/C SHIM COILS FOR GE MAGNETS ONLY.
 - FOR MOBILE SYSTEMS, SENSOR IS LOCATED IN CEILING BOX.
 - OPTIONAL RESISTIVE SHIM SUPPLY AND COIL.
 - DESIGN DIFFERENCES BETWEEN 1.5T & 1.0T SYSTEMS.
 - OPTIONS AVAILABLE FOR 1.5 T SYSTEMS ONLY.
 - HORIZON SYSTEMS HAVE LITRON HEAT EXCHANGER MR// AND CV// FIXED SITE SYSTEMS HAVE NESLAB WATER CHILLER. MOBILES HAVE E & W HEAT EXCHANGER.
 - SIGNA MR// AND CV// SYSTEMS MANUFACTURED AFTER JUNE 15, 2000 HAVE MICROPHONE IN THE HEAD CARRIAGE INSTEAD OF THE REAR END BELL.
 - FOR TSCC AND TGWC SEE WATER CHILLER.

REFER TO THE FOLLOWING FUNCTIONAL BLOCK DIAGRAM FOR ADDITIONAL INFORMATION:

NOTE:

- PATIENT HANDLING
- MGD/RRF/RF
- GRADIENT
- MAGNET AND CRYOGENS
- REFER TO SPECTROSCOPY MANUALS
- OXYGEN MONITOR SUBSYSTEM (DIR. 15336).

OVERALL SYSTEM - EXCITE 2
(ONLY MAIN COMMUNICATION LINES SHOWN)

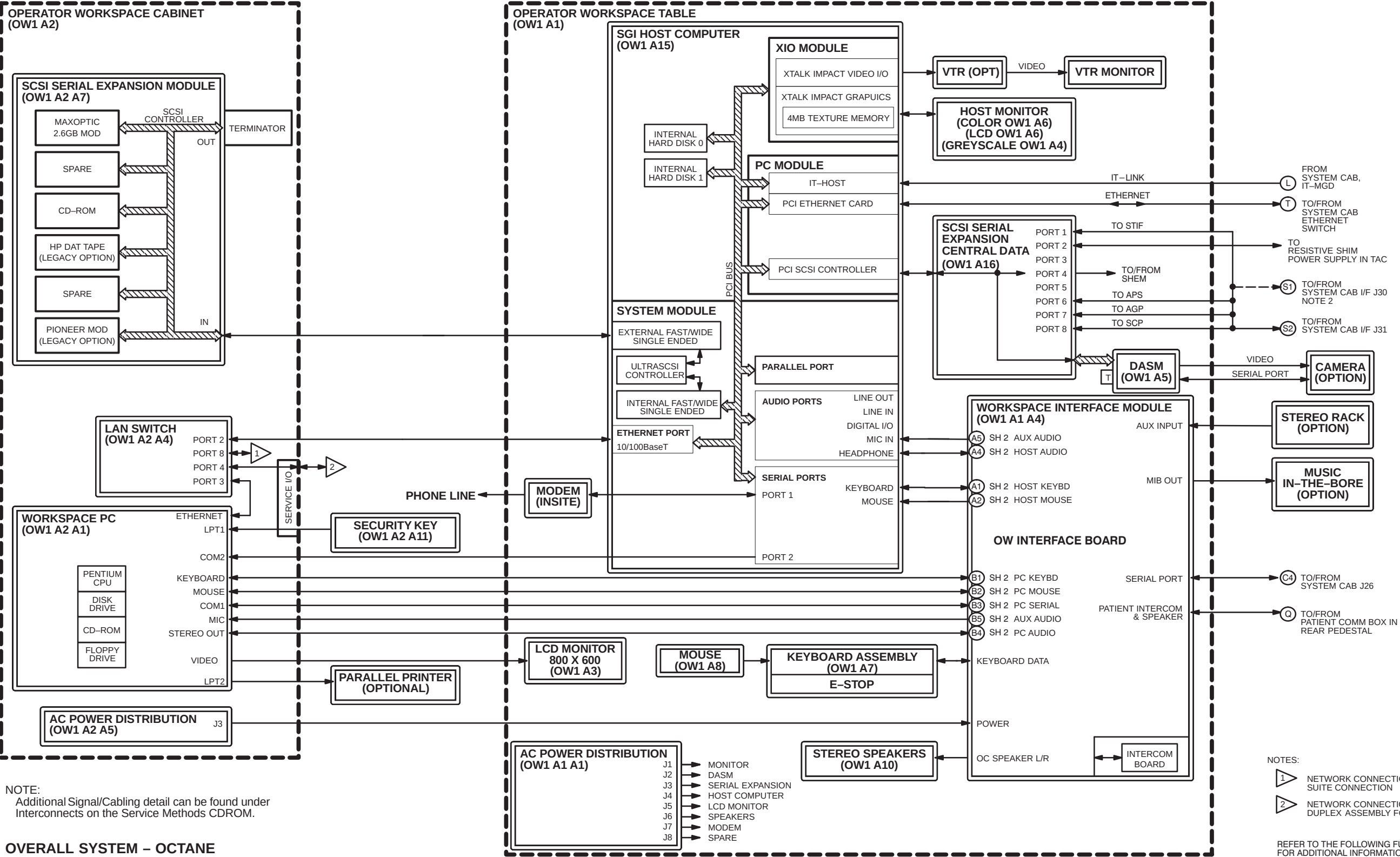
SHEET 4 OF 4

OPERATOR WORKSPACE

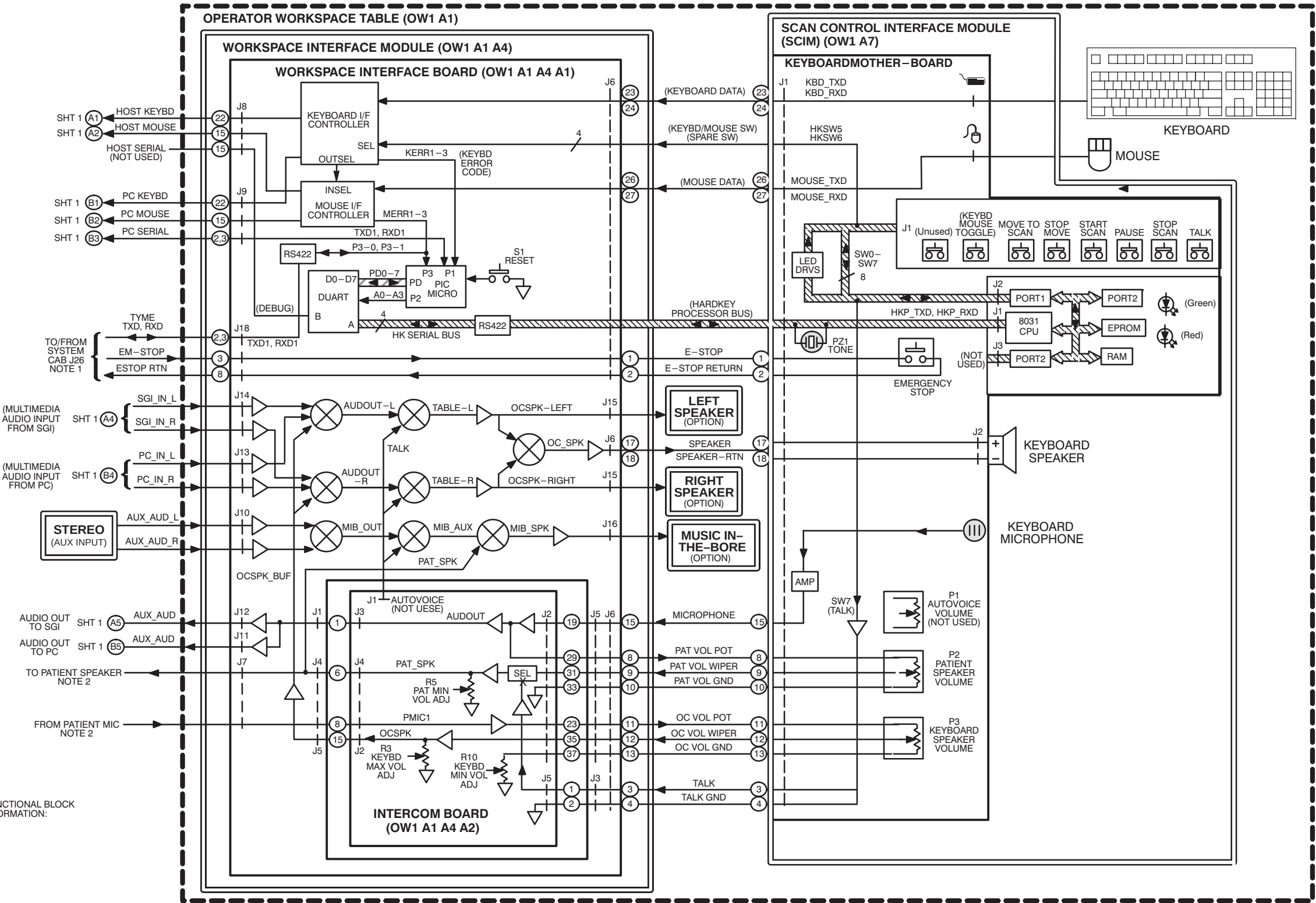
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2 – OPERATOR WORKSPACE – LINUX	2-1

NOTE:
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.



OVERALL SYSTEM – OCTANE
(OPERATOR WORKSPACE CABINET/TABLE – OCTANE)
SHEET 1 OF 2

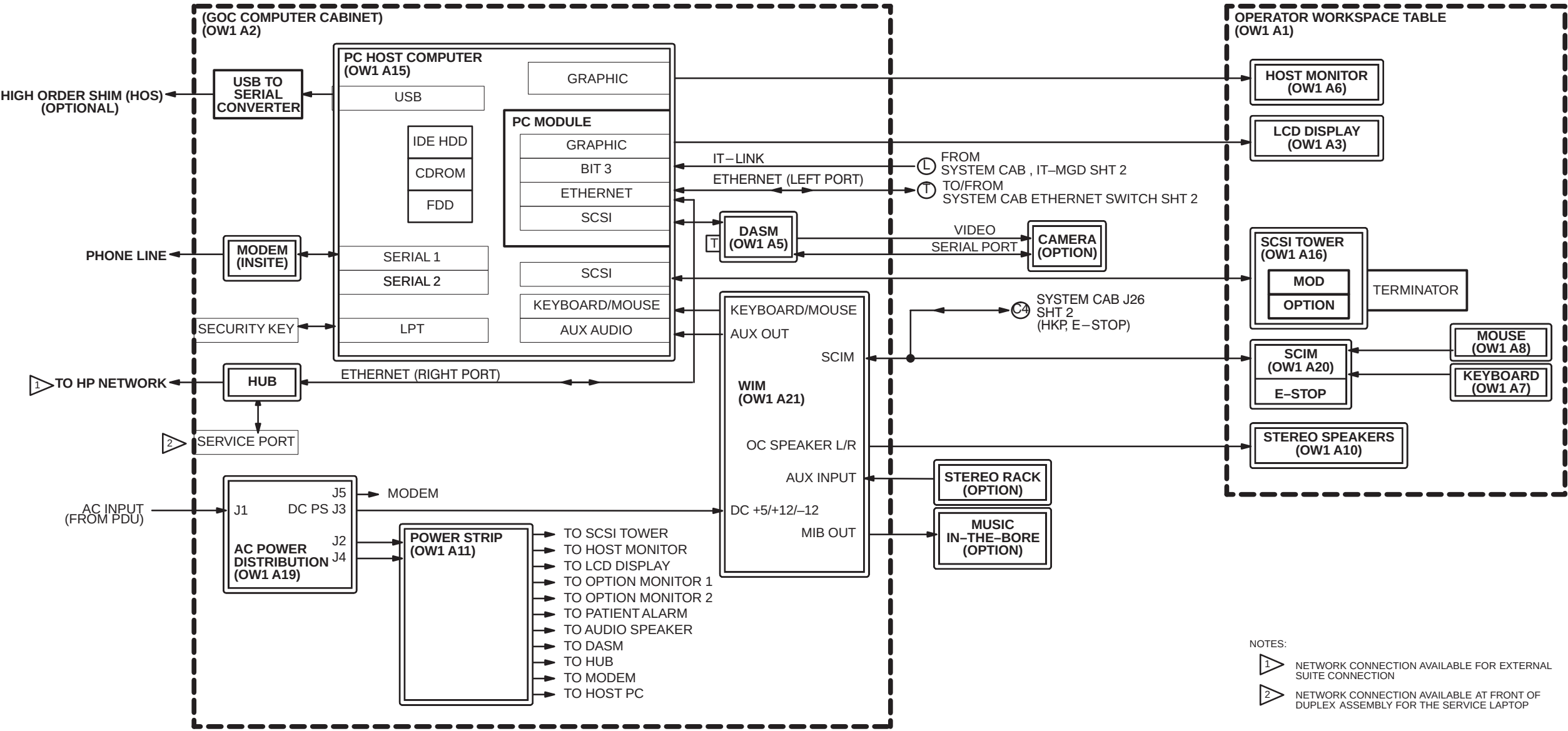


REFER TO THE FOLLOWING FUNCTIONAL BLOCK
DIAGRAM FOR ADDITIONAL INFORMATION:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF
3) OVERALL SYSTEM

OVERALL SYSTEM – OCTANE
(OPERATOR WORKSPACE CABINET/TABLE – OCTANE)
SHEET 2 OF 2

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NOTE:
ADDITIONAL SIGNAL/CABLING DETAIL CAN BE FOUND UNDER
INTERCONNECTS ON THE SERVICE METHODS CDROM.

OVERALL SYSTEM – LINUX
(OPERATOR WORKSPACE CABINET/TABLE – LINUX)
SHEET 1 OF 2

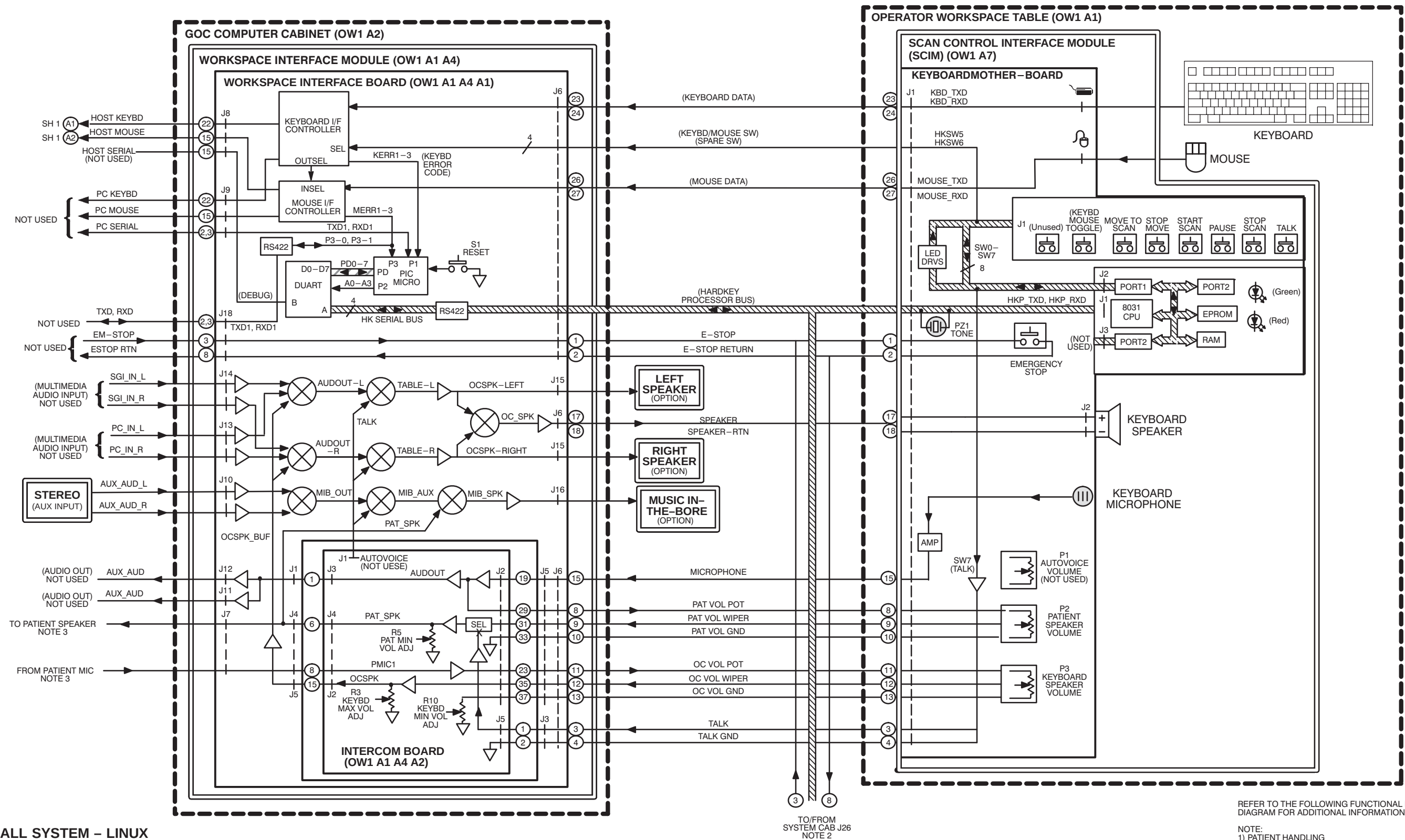
NOTES:

1 NETWORK CONNECTION AVAILABLE FOR EXTERNAL
SUITE CONNECTION

2 NETWORK CONNECTION AVAILABLE AT FRONT OF
DUPLEX ASSEMBLY FOR THE SERVICE LAPTOP

REFER TO THE FOLLOWING FUNCTIONAL BLOCK DIAGRAM
FOR ADDITIONAL INFORMATION:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF



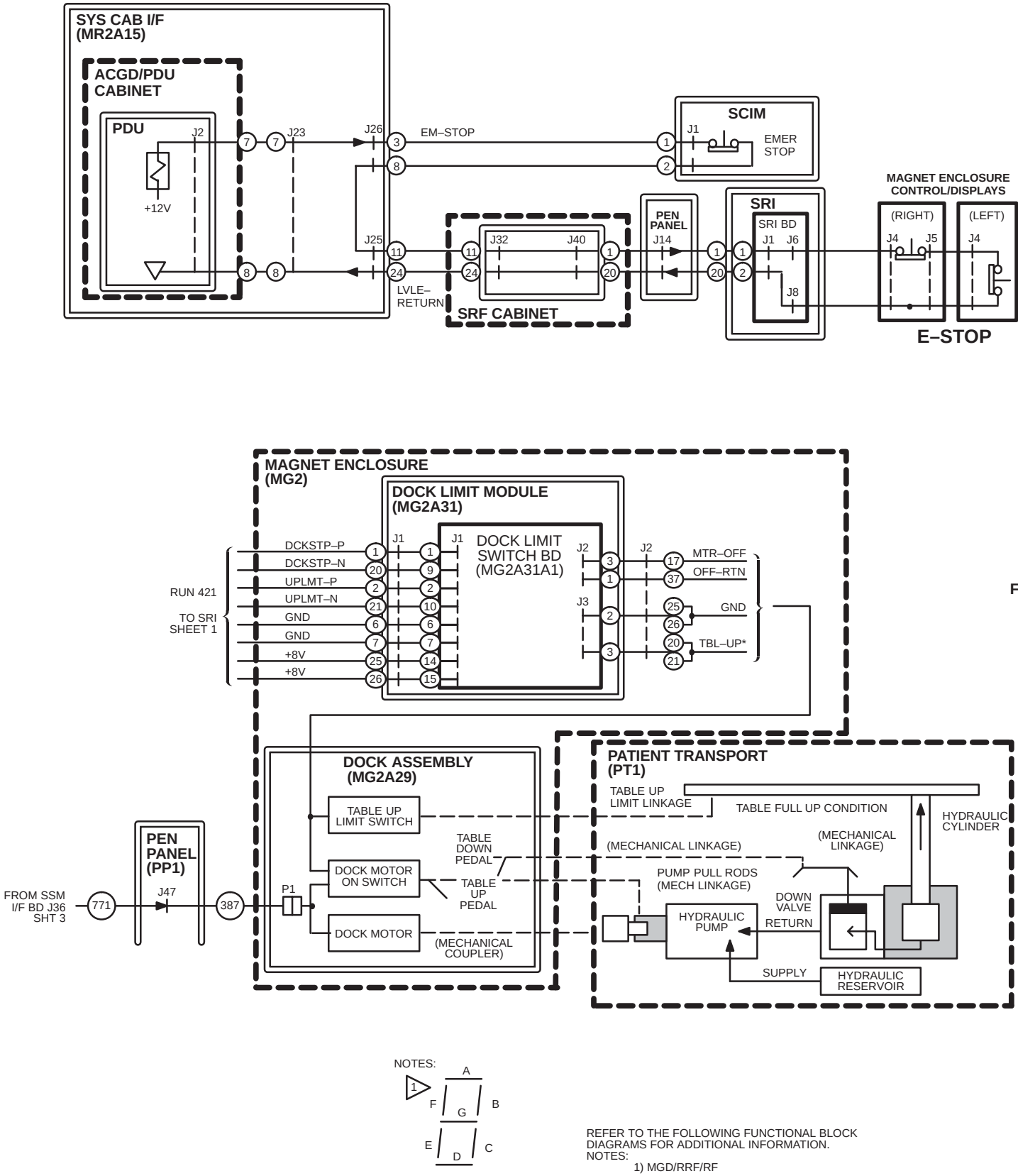
SYSTEM

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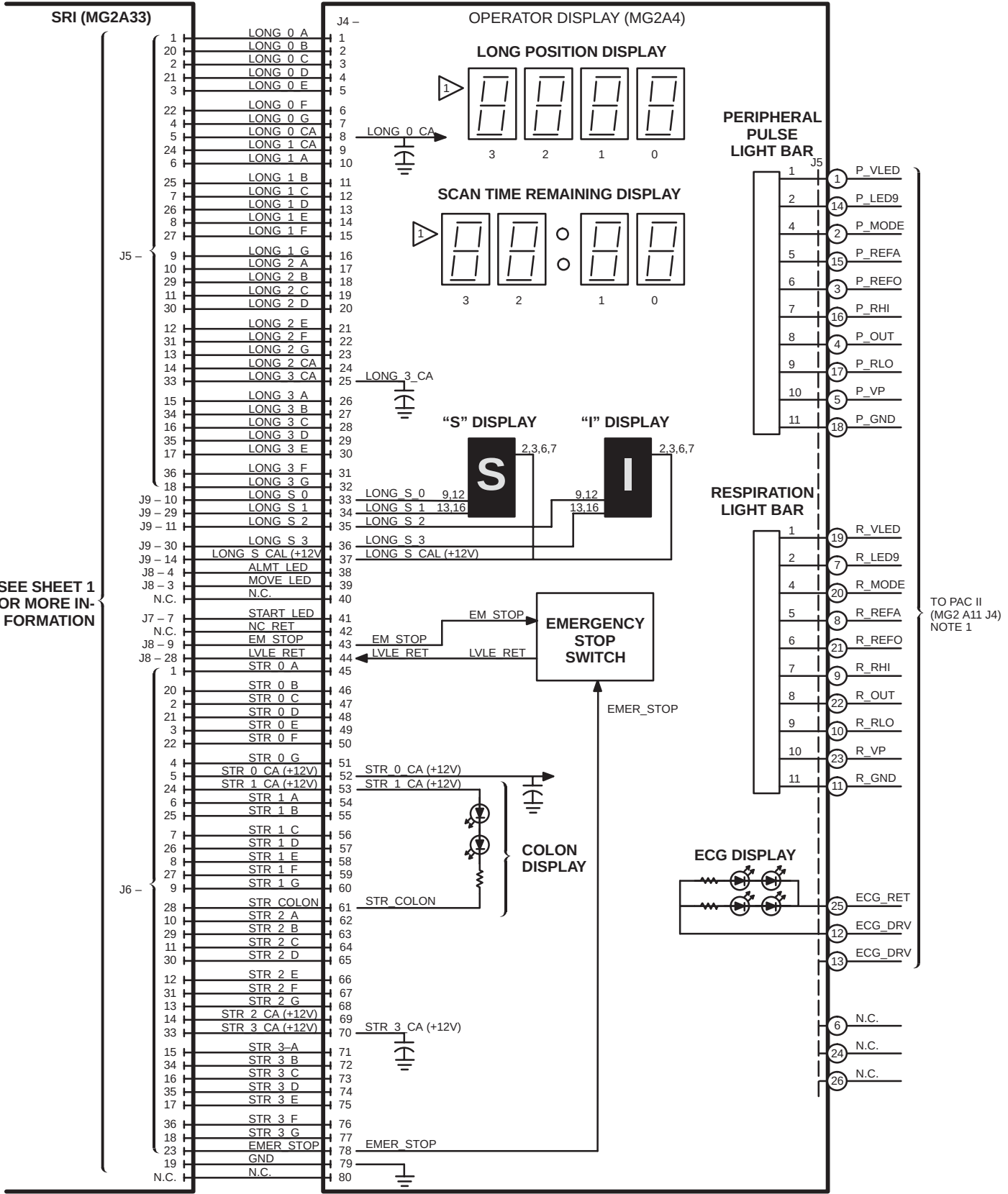
<u>SECTION</u>	<u>PAGE</u>
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2 – PATIENT HANDLING – LINUX	2-1

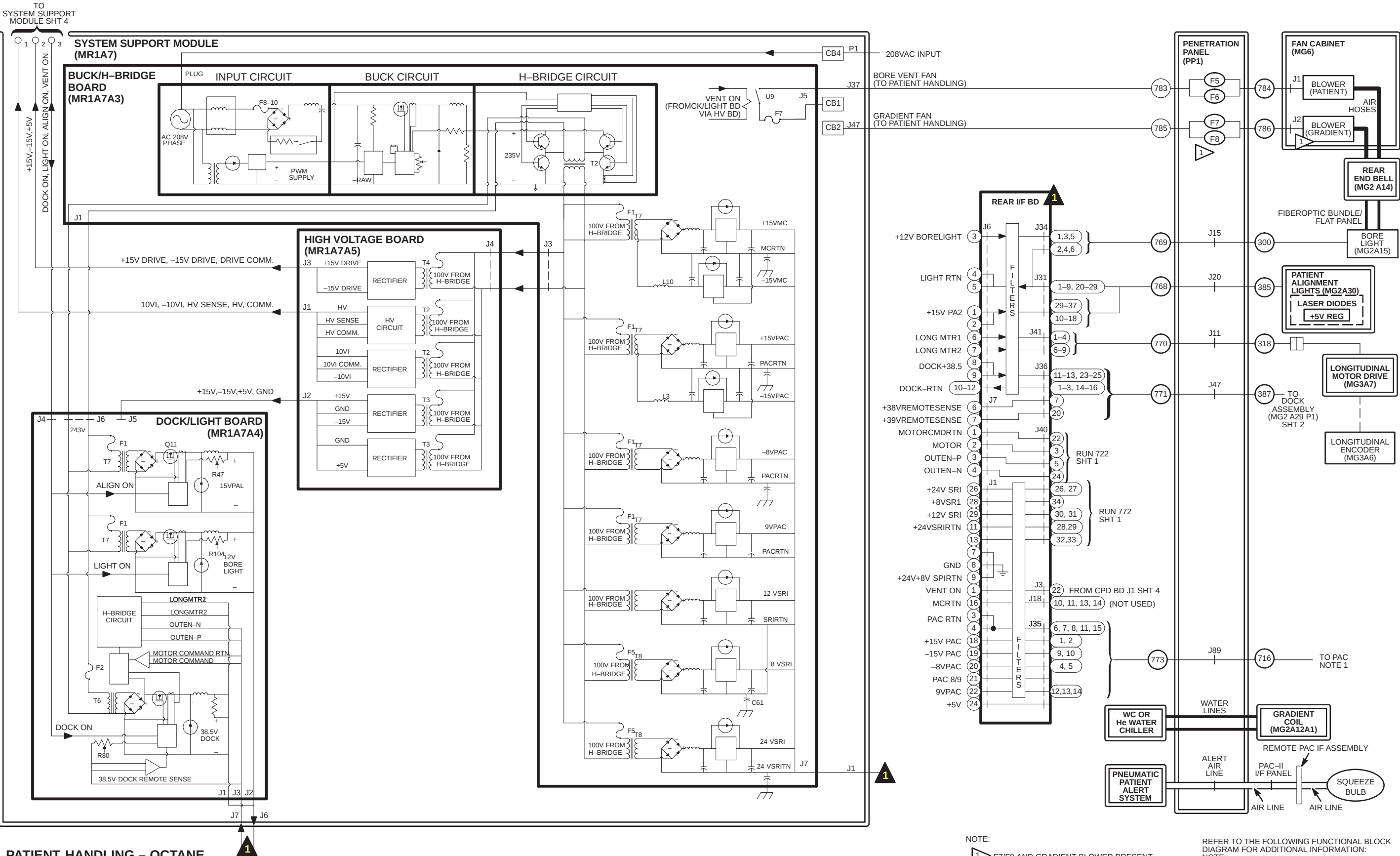
NOTE:
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

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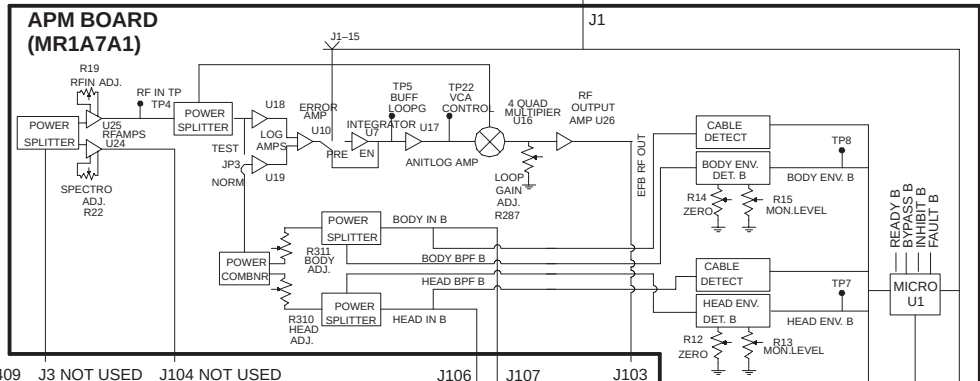
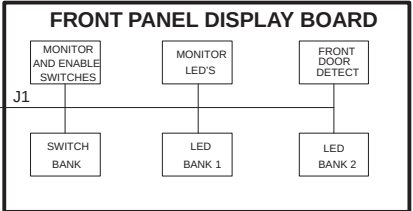
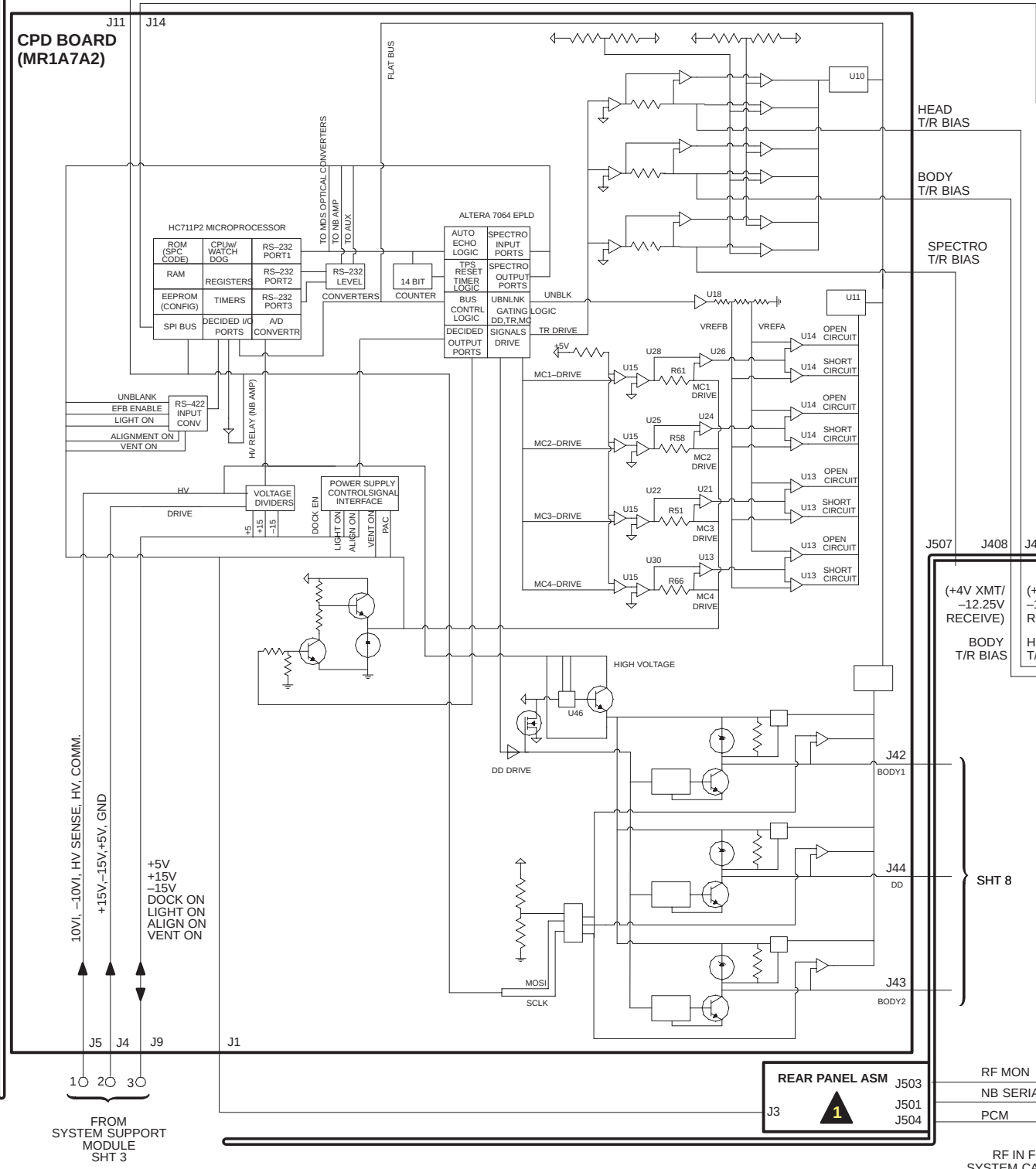
PATIENT HANDLING – OCTANE
SHEET 2 OF 4





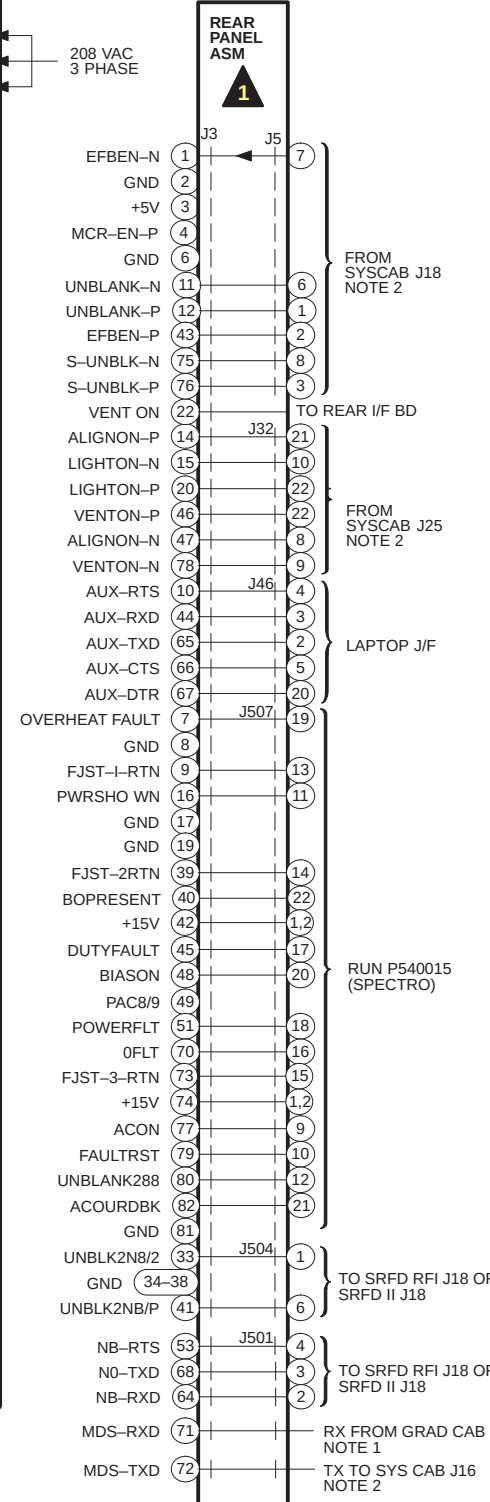
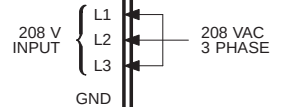
SYSTEM SUPPORT MODULE
(MR1A7)

CPD BOARD
(MR1A7A2)



SRFD OR SRFD II CABINET

SEE MGD/RRF/RFSUBSYSTEM SECTION:
SRFD POWER CABINET (SHT 6) OR
SRFD II POWER CABINET (SHT 7A)

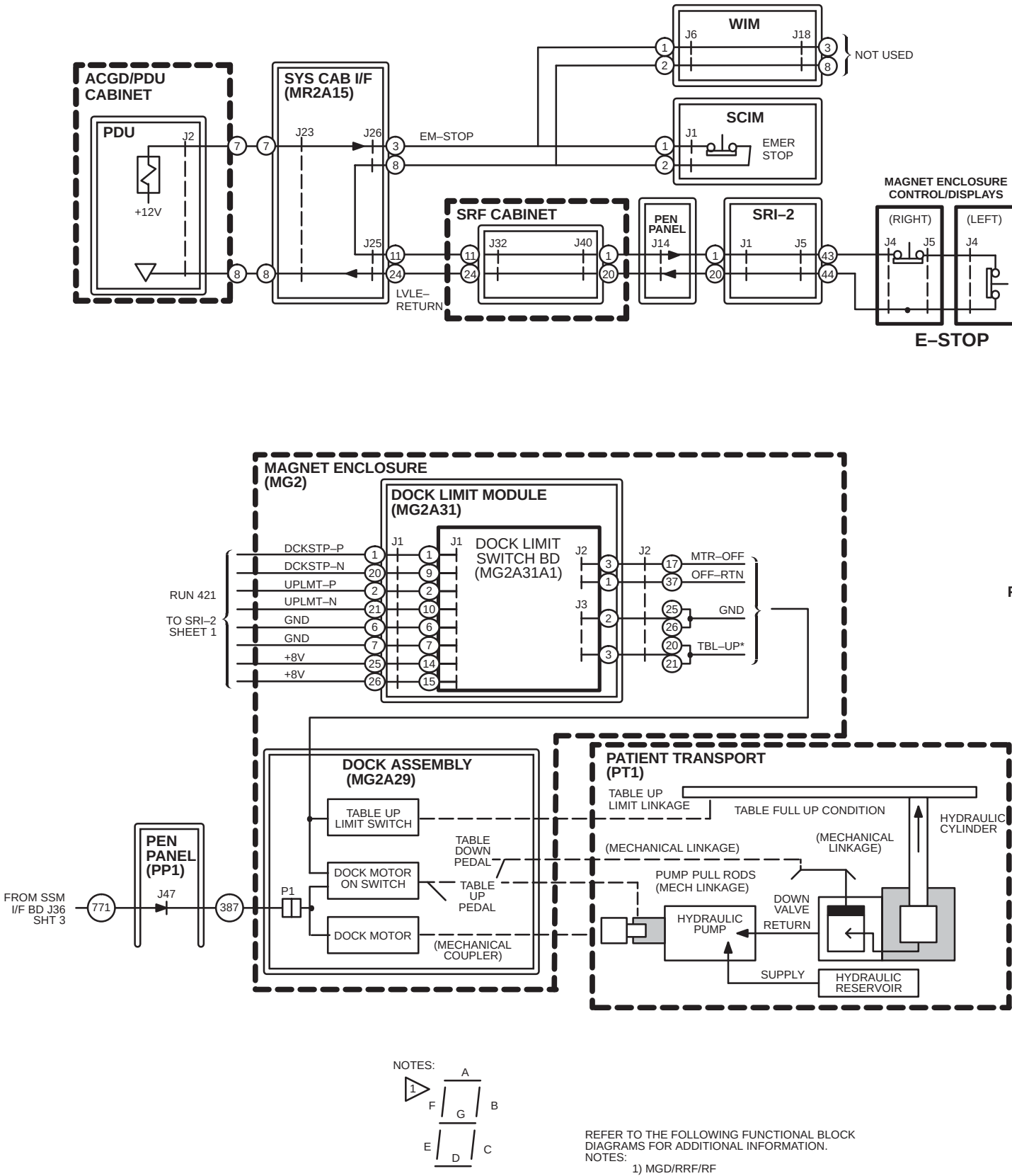


REFER TO THE FOLLOWING FUNCTIONAL BLOCK
DIAGRAM FOR ADDITIONAL INFORMATION:

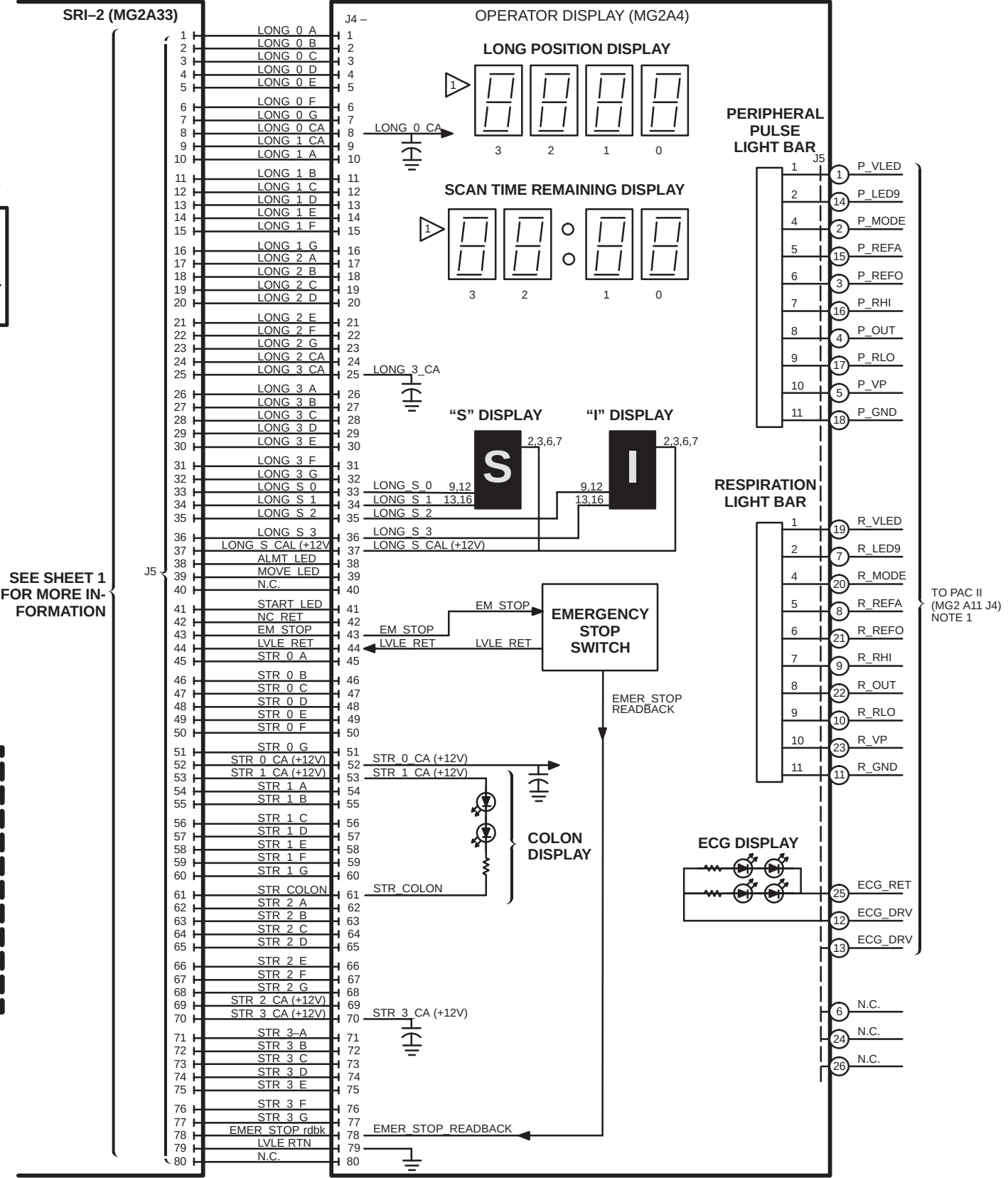
NOTE:
1) GRADIENT
2) MGD/RRF/RF

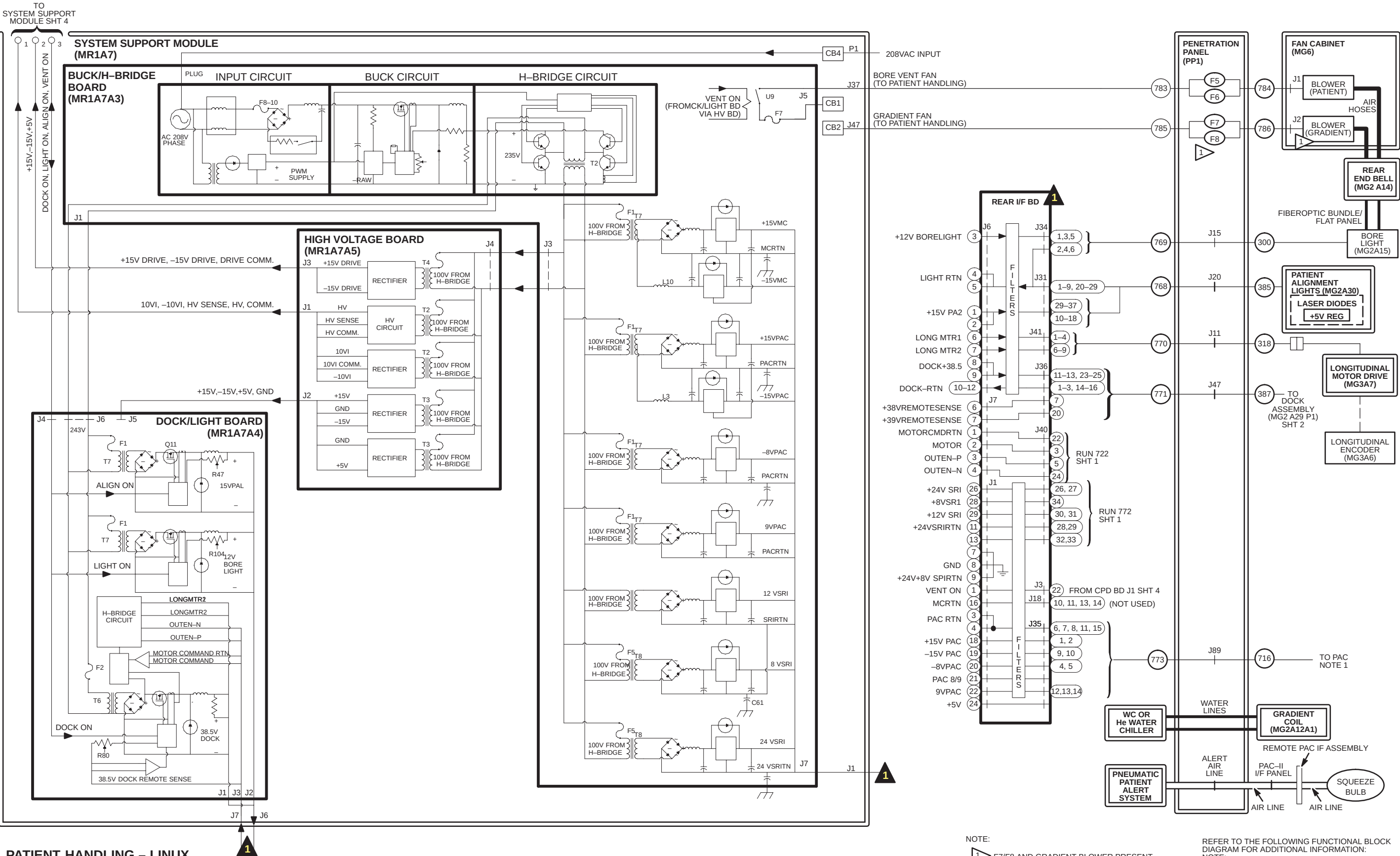


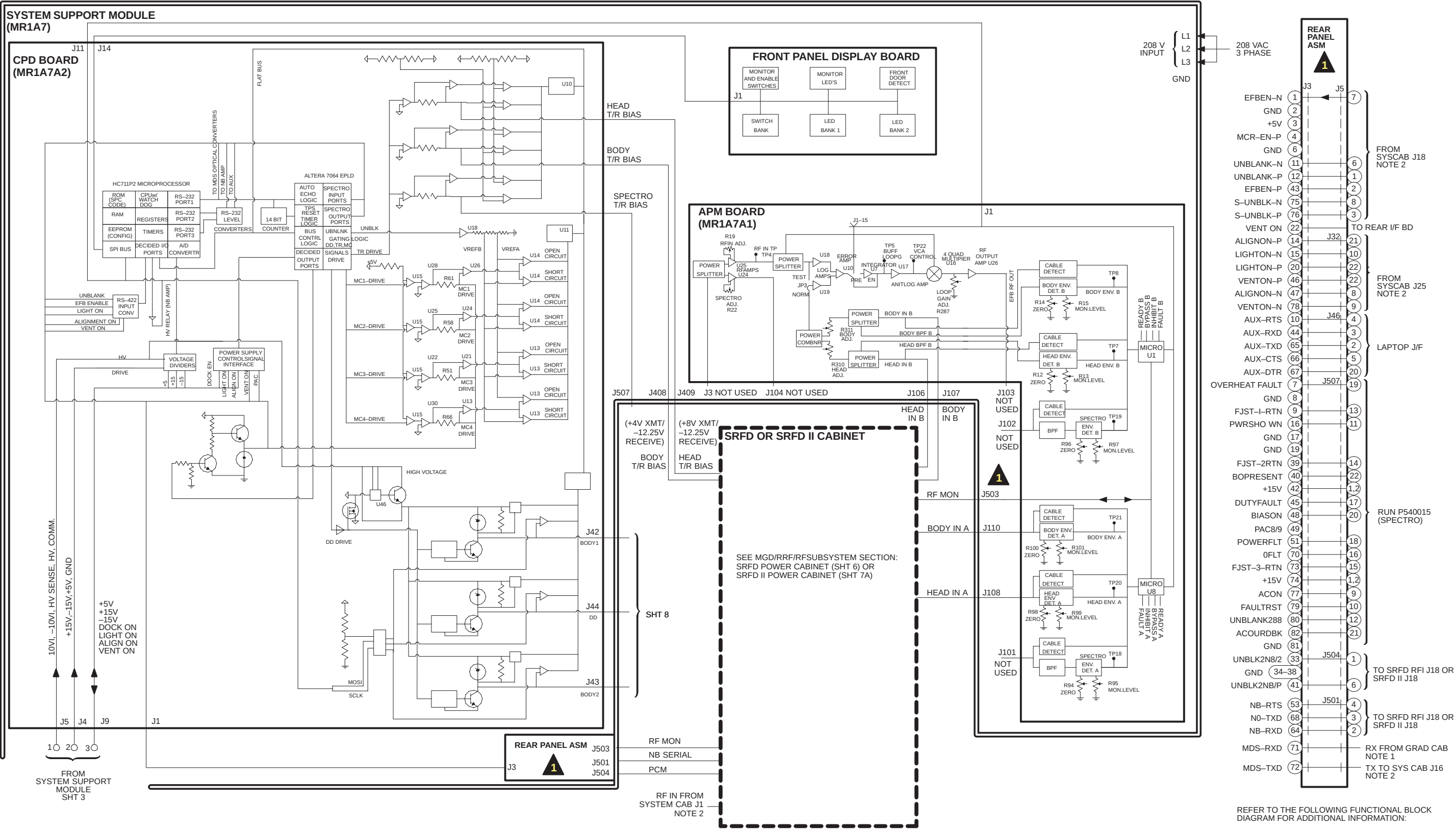
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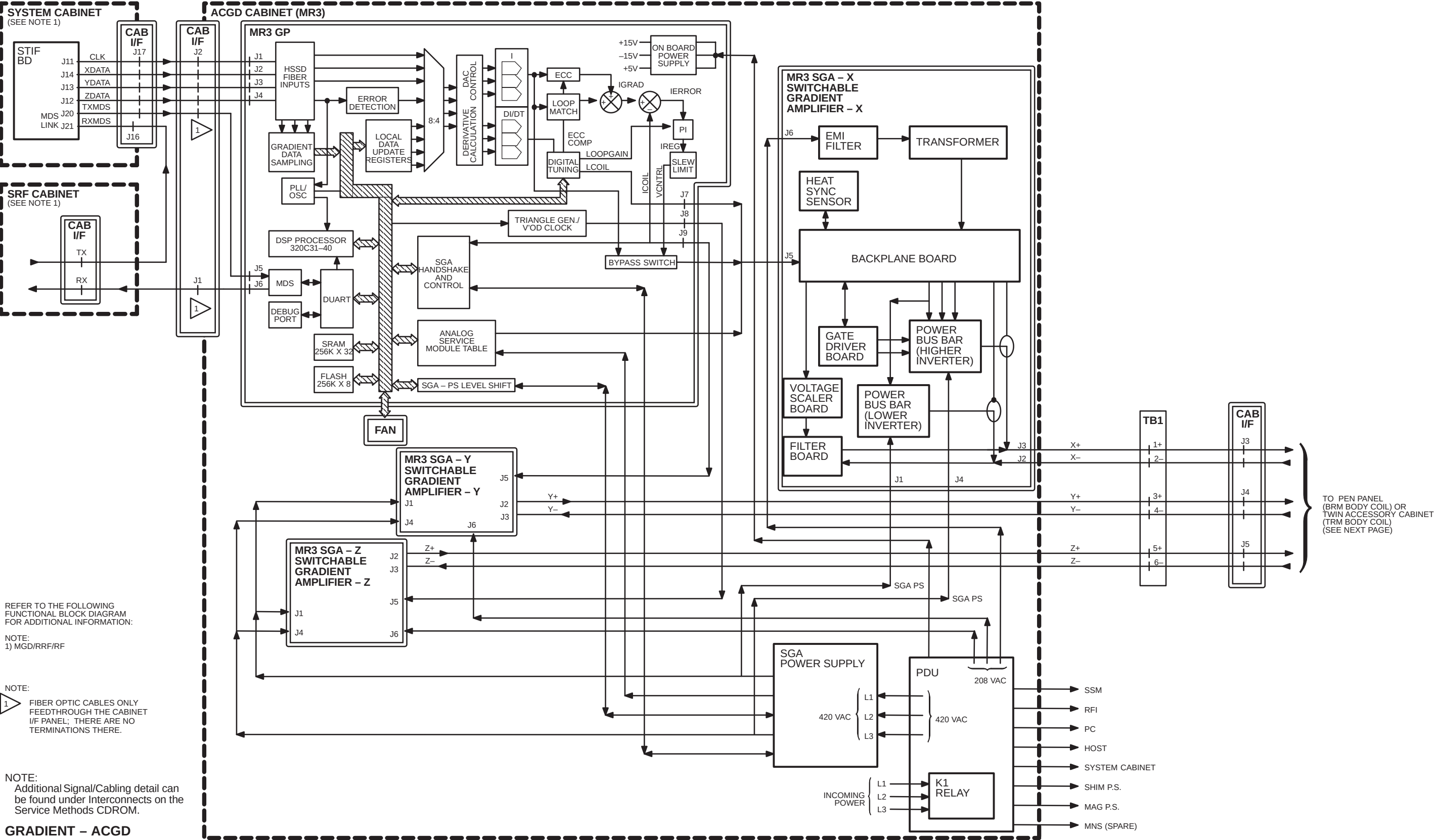


PATIENT HANDLING - EXCITE 2
SHEET 2 OF 4











MGD/RRF/RF (1.5T)

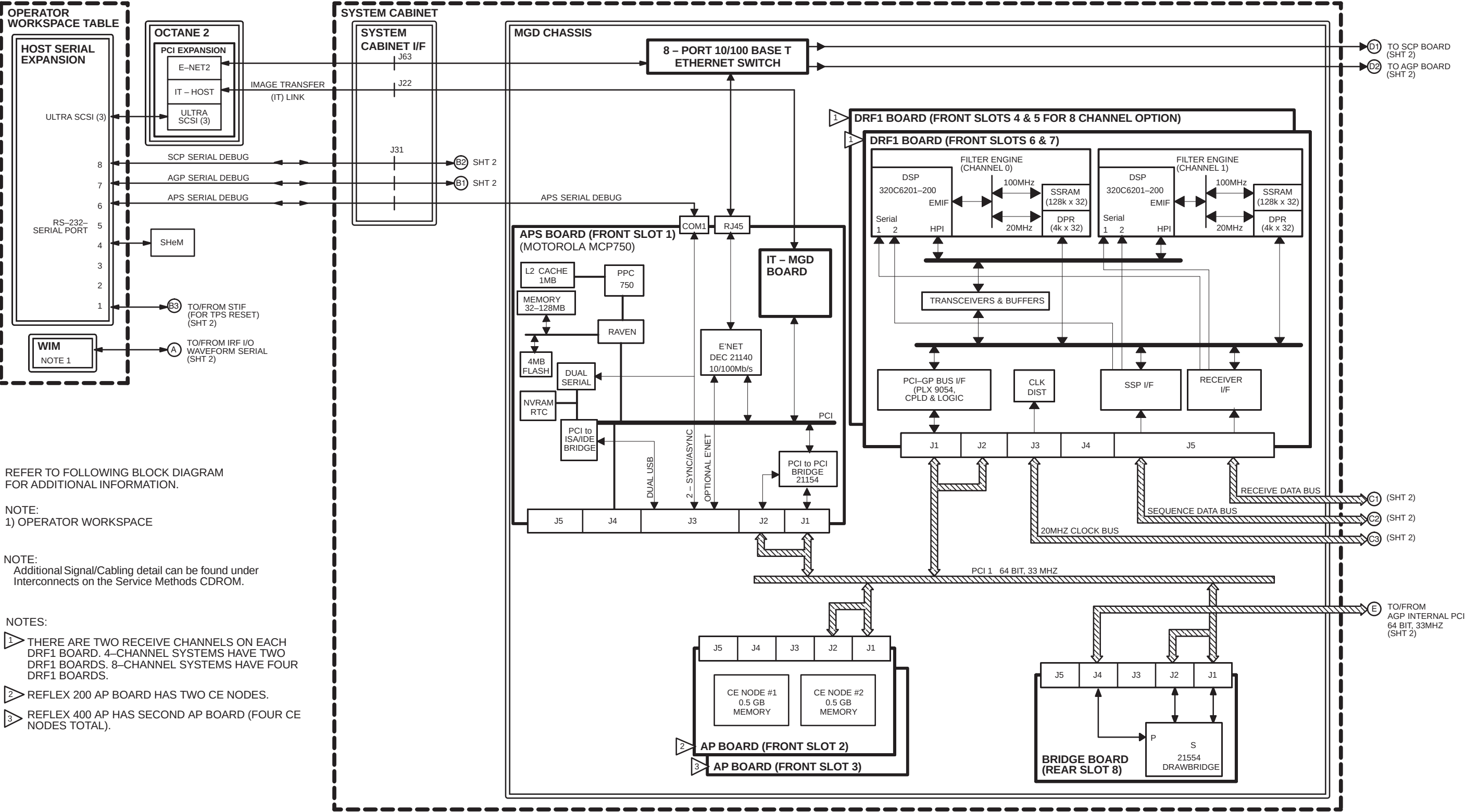
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DRIVER MODULE	1–7
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NOTE:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

SECTION 1 – MGD/RRF/RF SUBSYSTEM (1.5T)

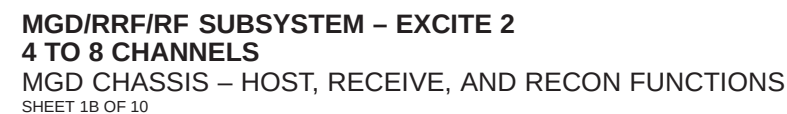


REFER TO FOLLOWING BLOCK DIAGRAM FOR ADDITIONAL INFORMATION.

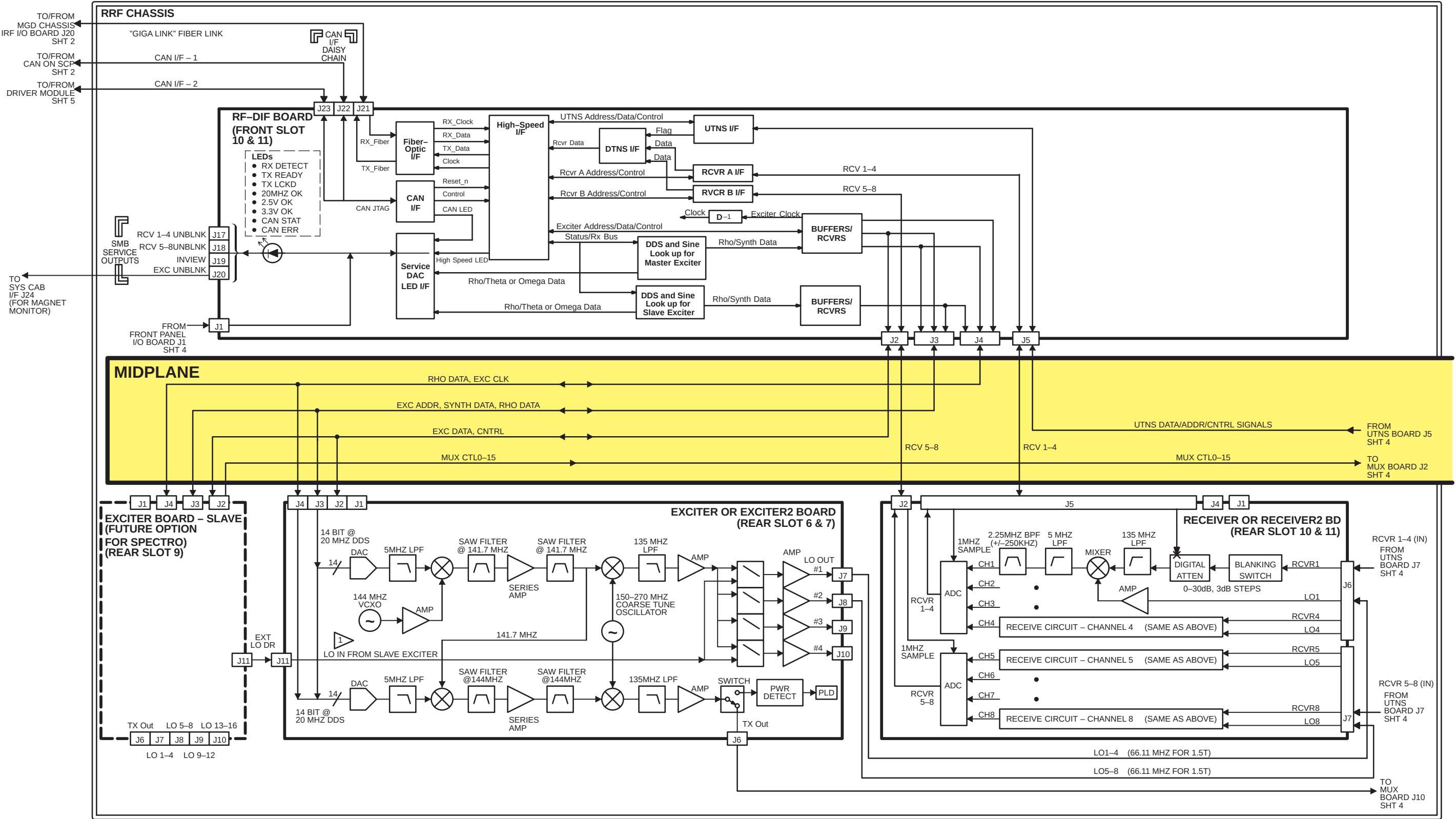
NOTE:
1) OPERATOR WORKSPACE

NOTE:
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

- NOTES:
- 1 THERE ARE TWO RECEIVE CHANNELS ON EACH DRF1 BOARD. 4-CHANNEL SYSTEMS HAVE TWO DRF1 BOARDS. 8-CHANNEL SYSTEMS HAVE FOUR DRF1 BOARDS.
 - 2 REFLEX 200 AP BOARD HAS TWO CE NODES.
 - 3 REFLEX 400 AP HAS SECOND AP BOARD (FOUR CE NODES TOTAL).

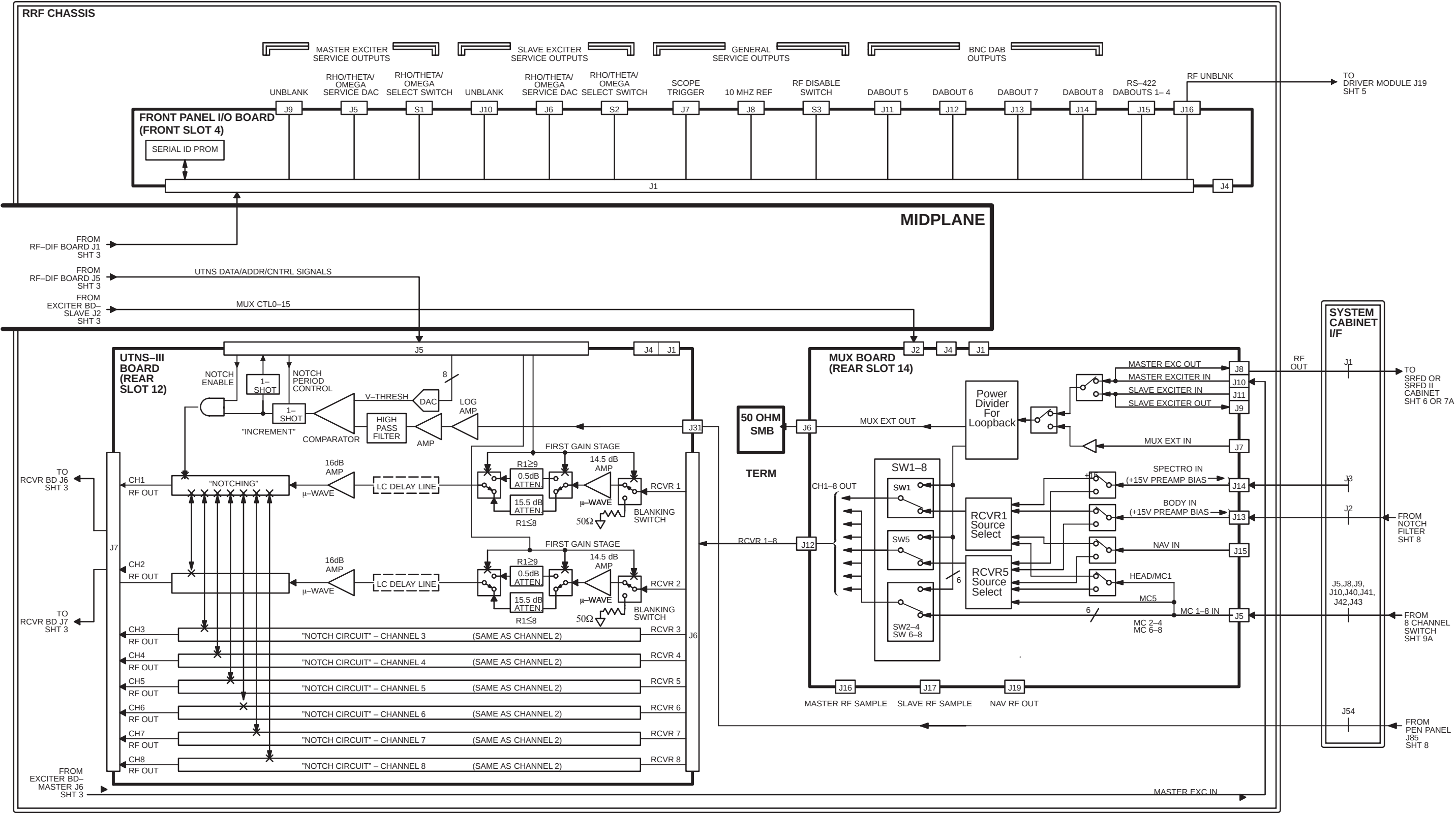






MGD/RRF/RF SUBSYSTEM
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 1 OF 2
SHEET 3 OF 10

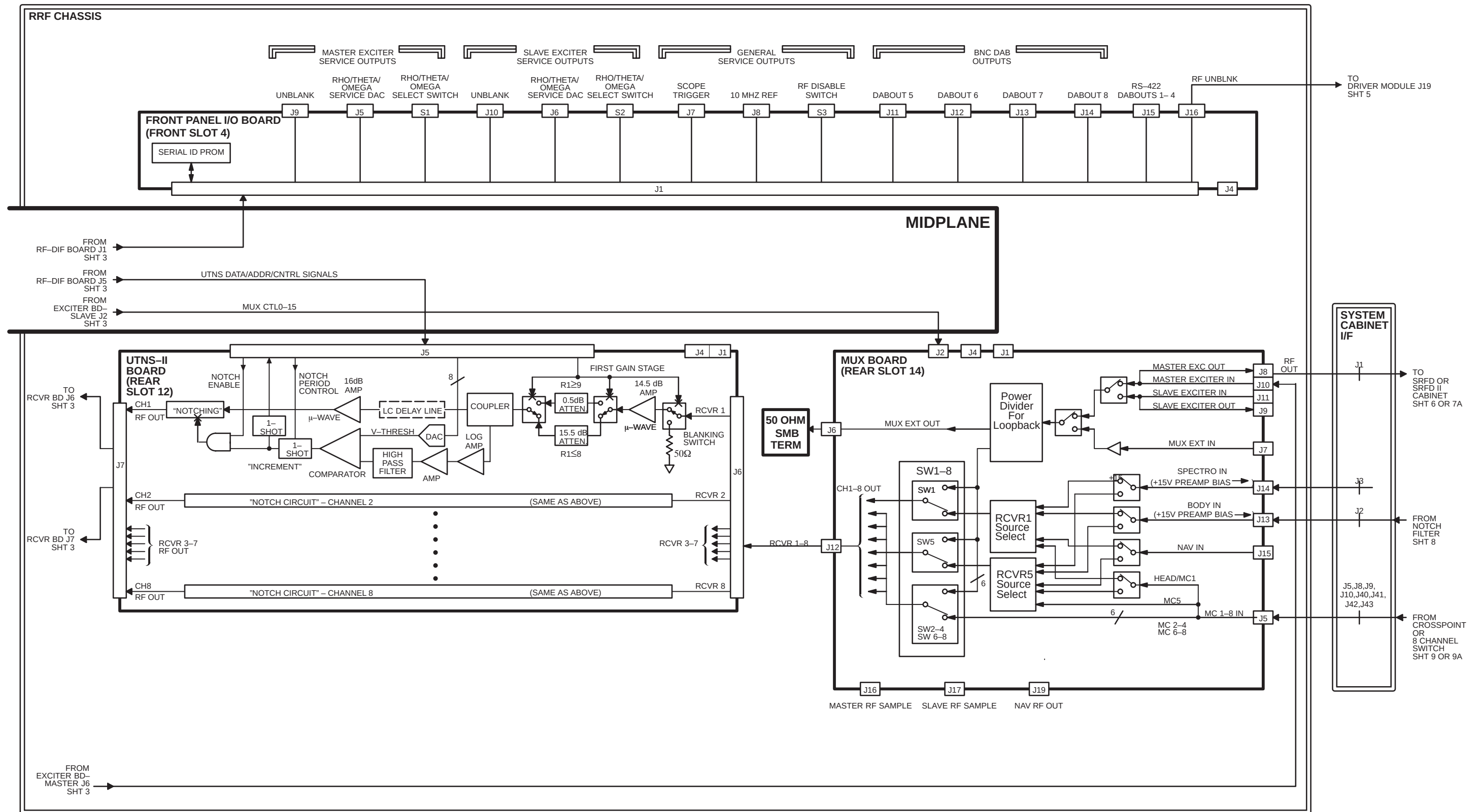
NOTE:
1 SLAVE EXCITER IS A FUTURE OPTION FOR SPECTRO.



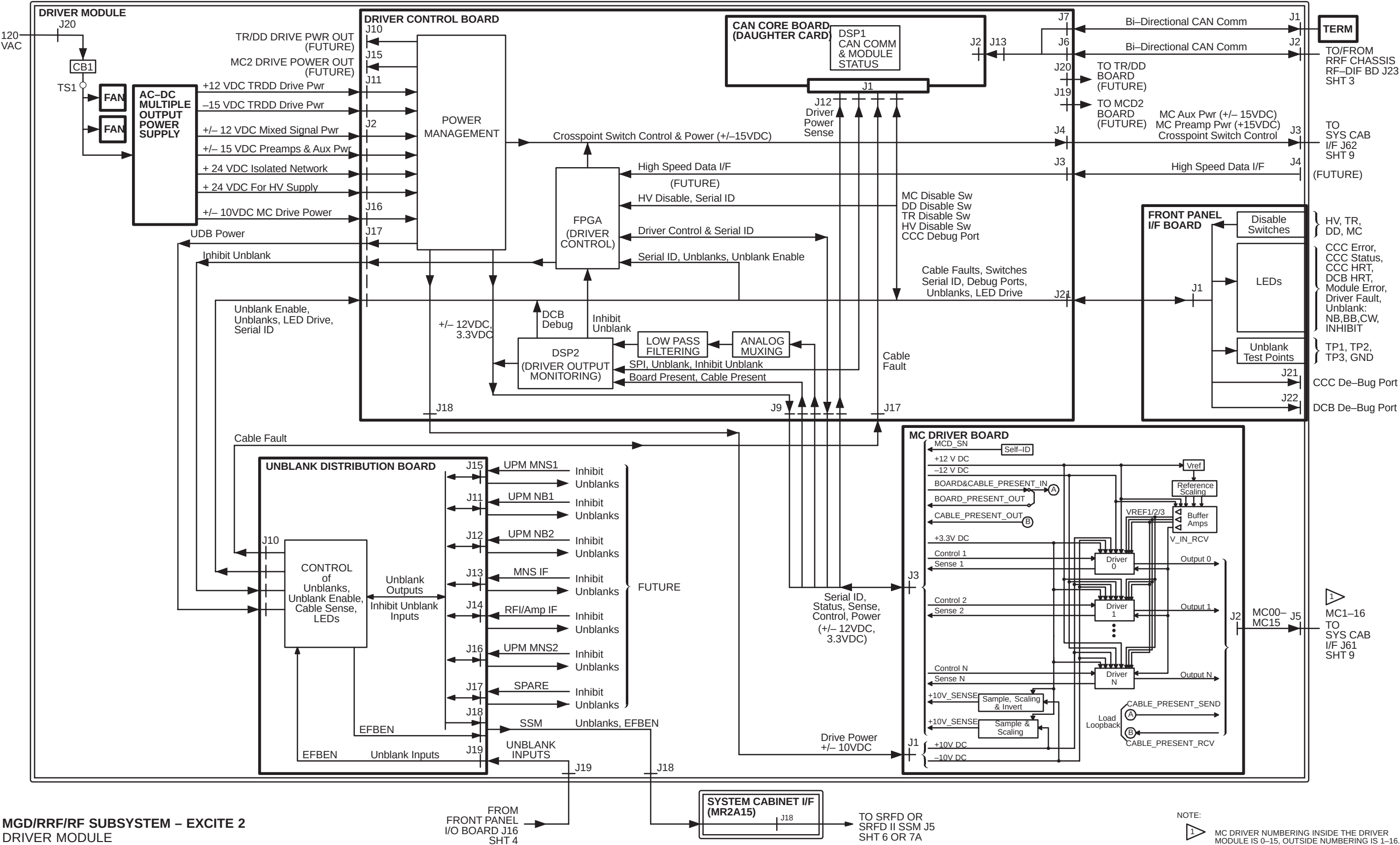
MGD/RRF/RF SUBSYSTEM with UTNS3

RFR CHASSIS INTERCONNECT BLOCK DIAGRAM 2A OF 2

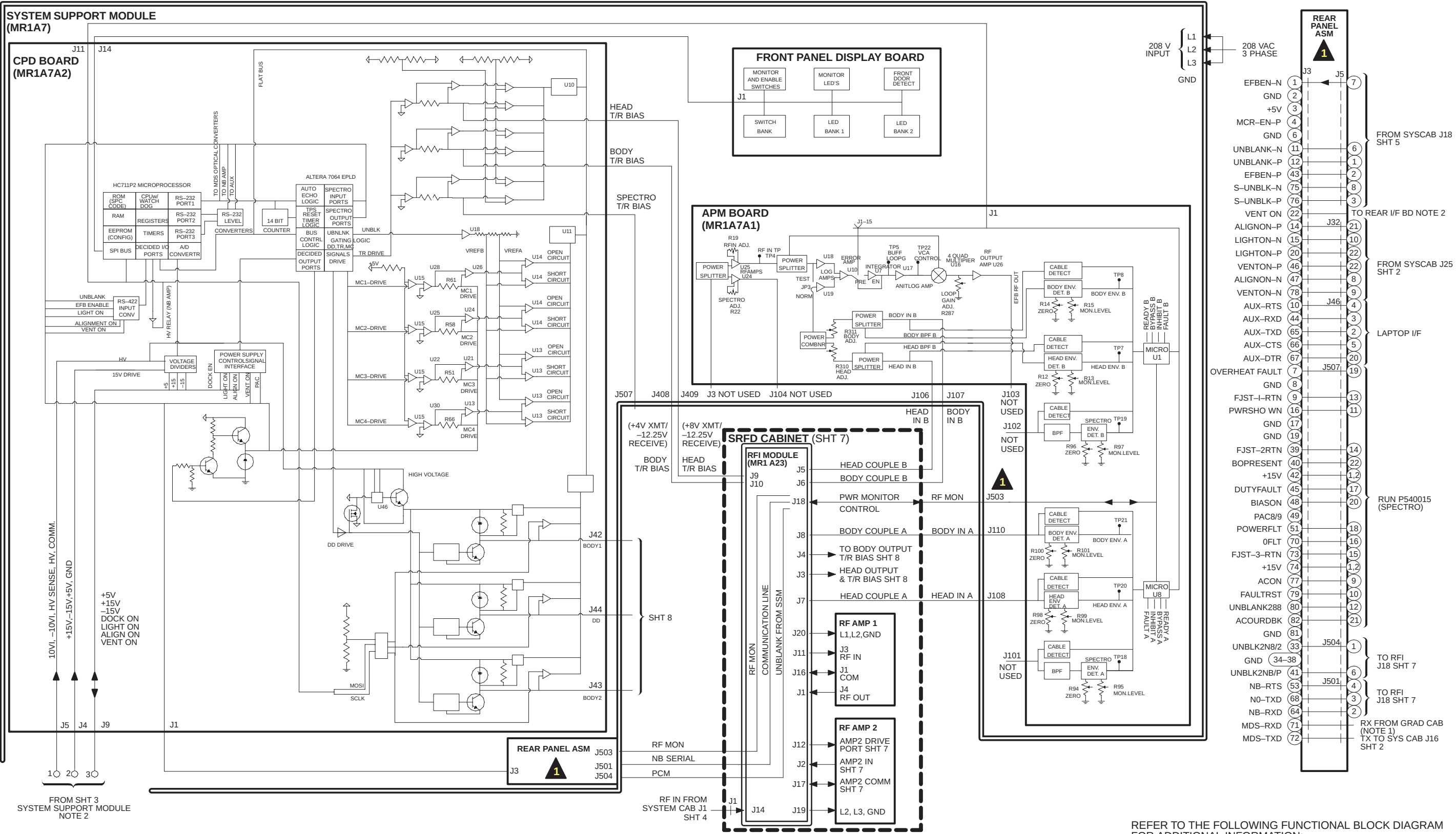
SHEET 4A OF 10

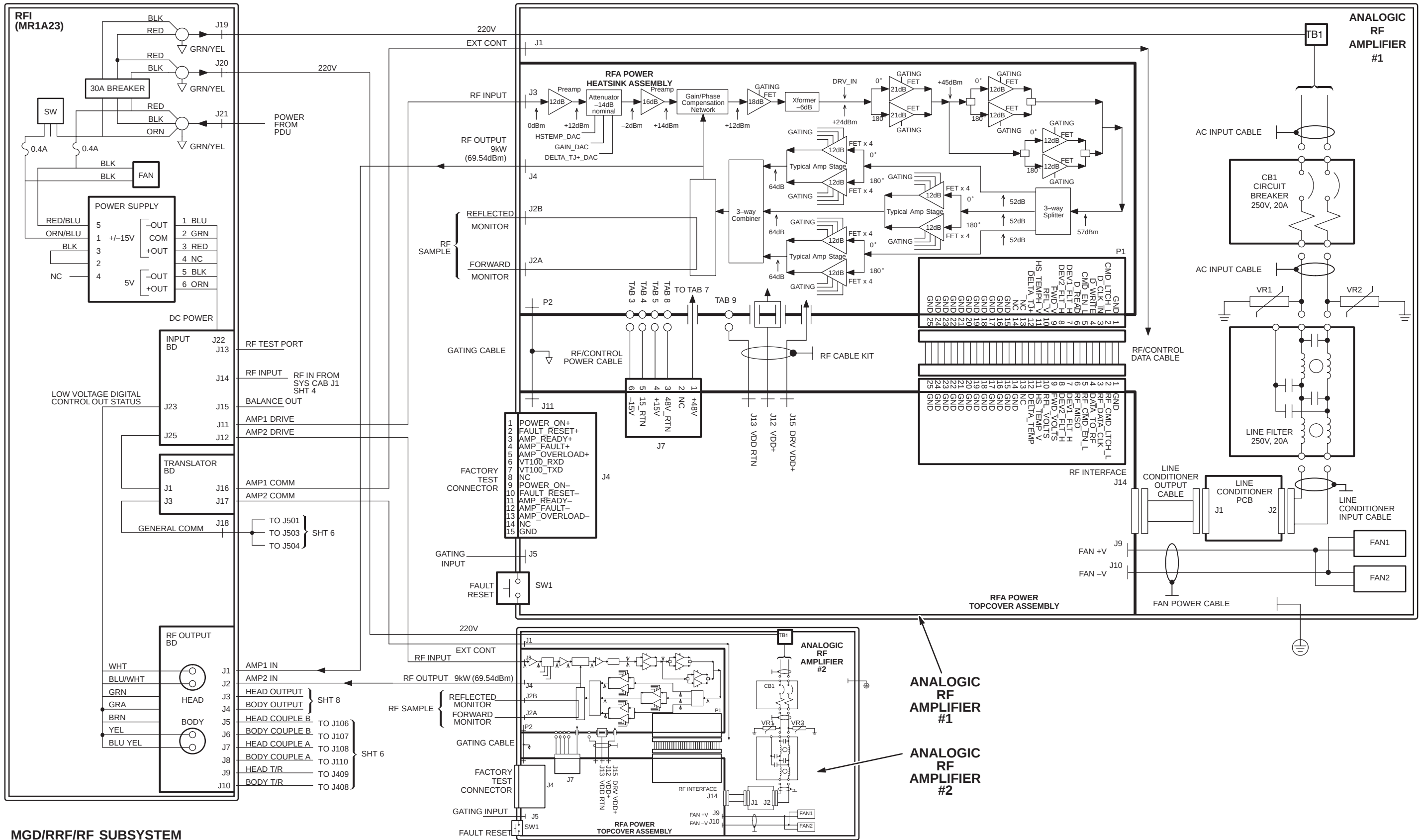


MGD/RRF/RF SUBSYSTEM with UTNS2 – EXCITE 2
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2B OF 2
 SHEET 4B OF 10

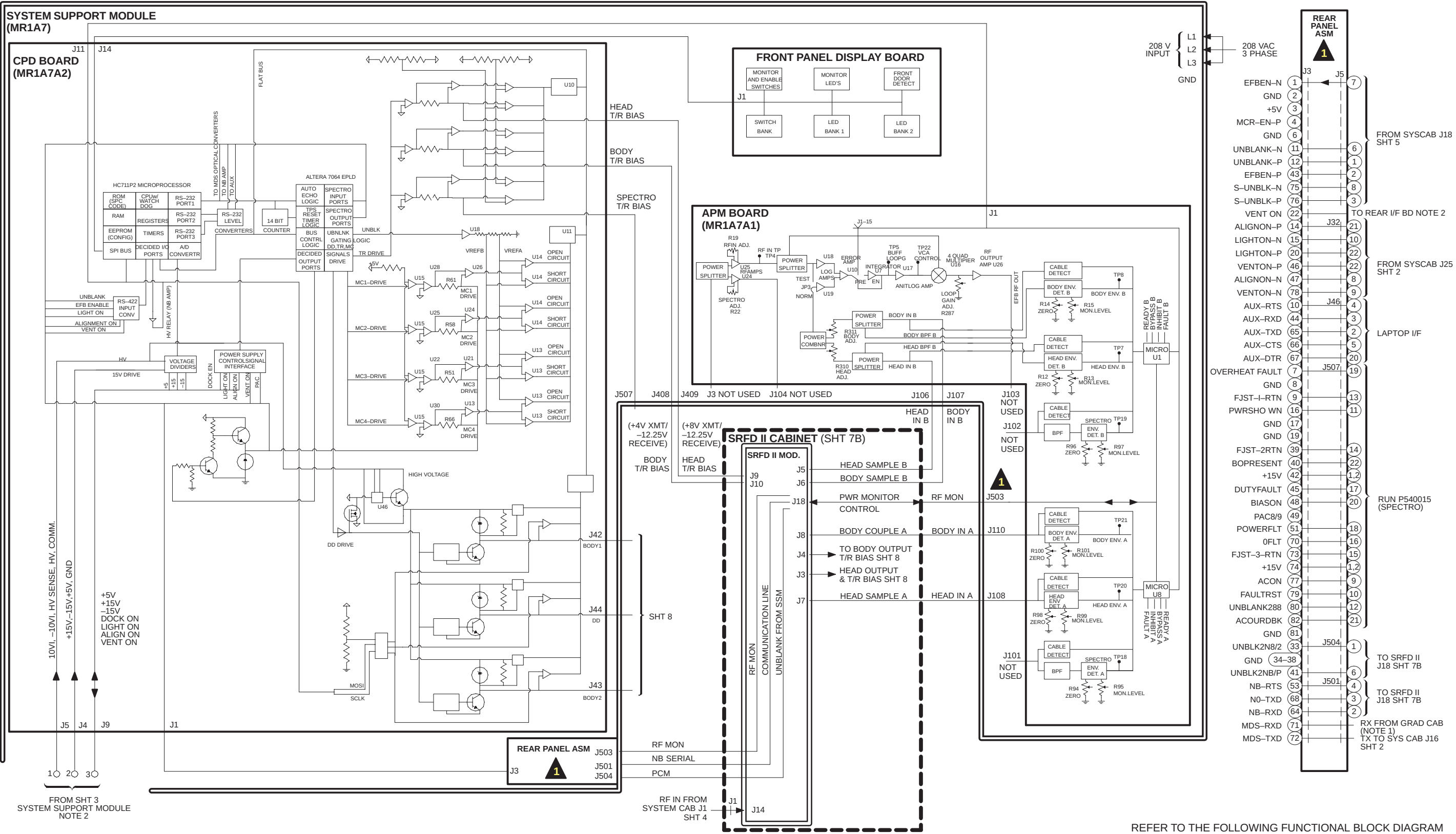


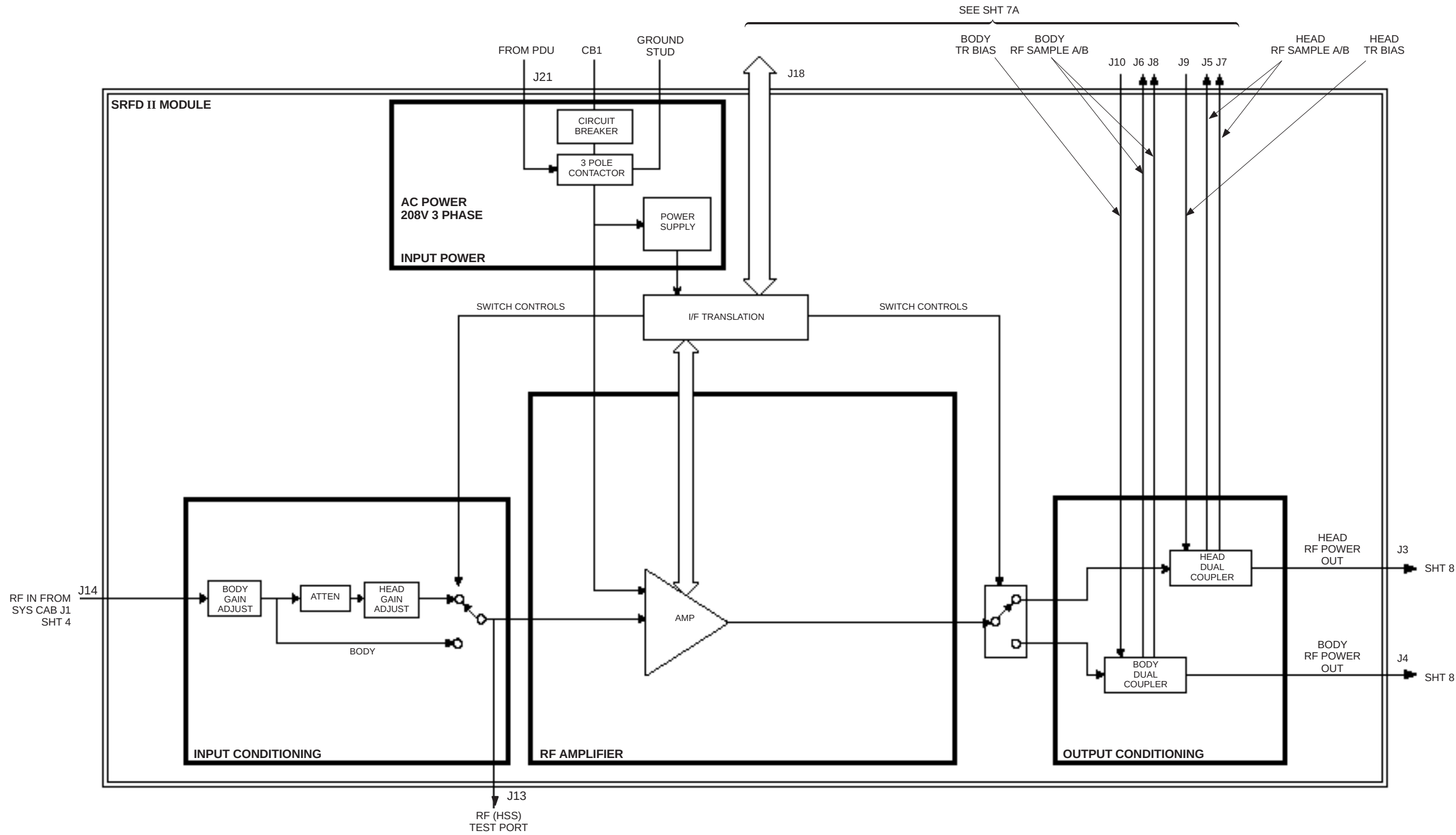
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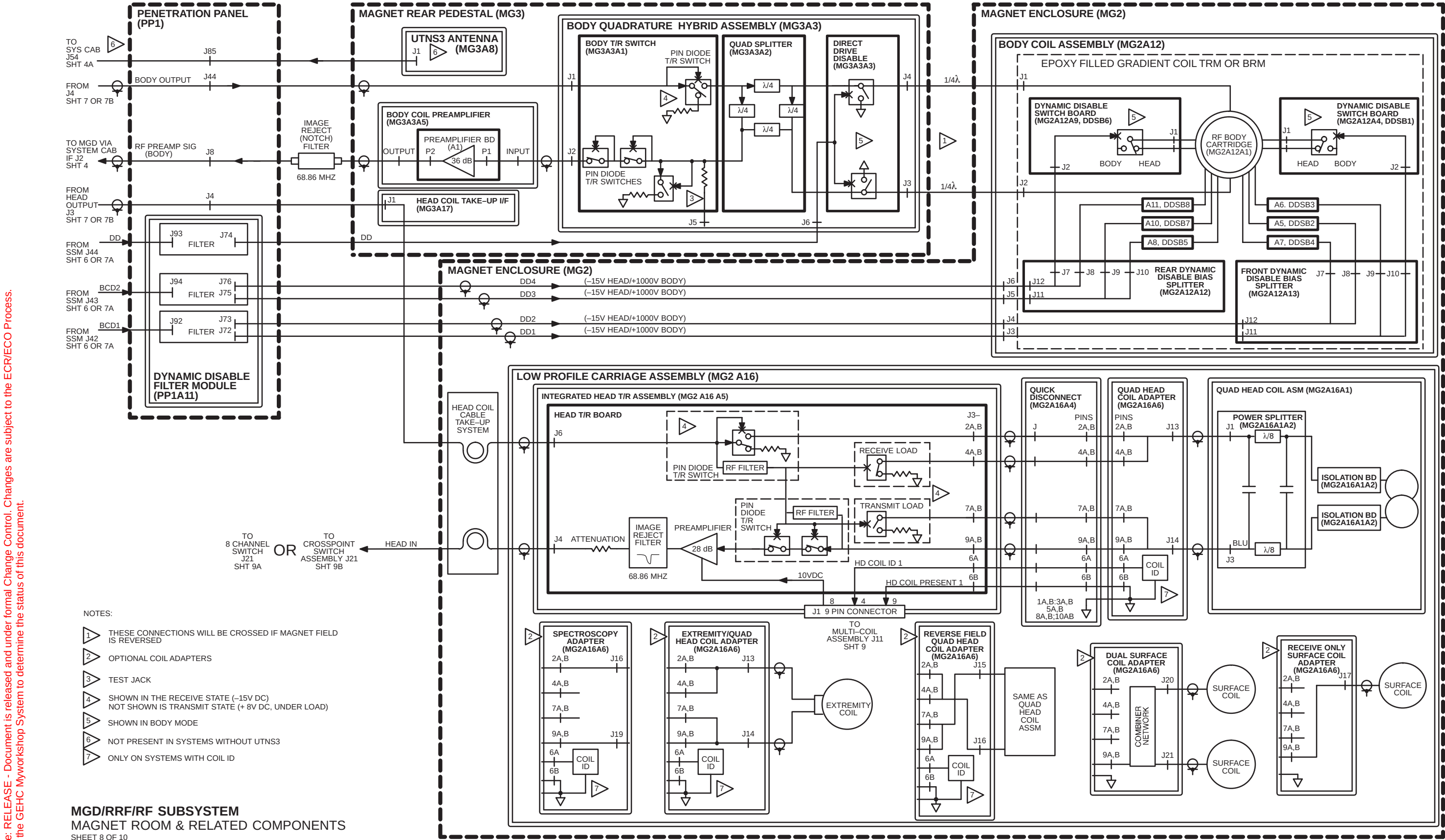


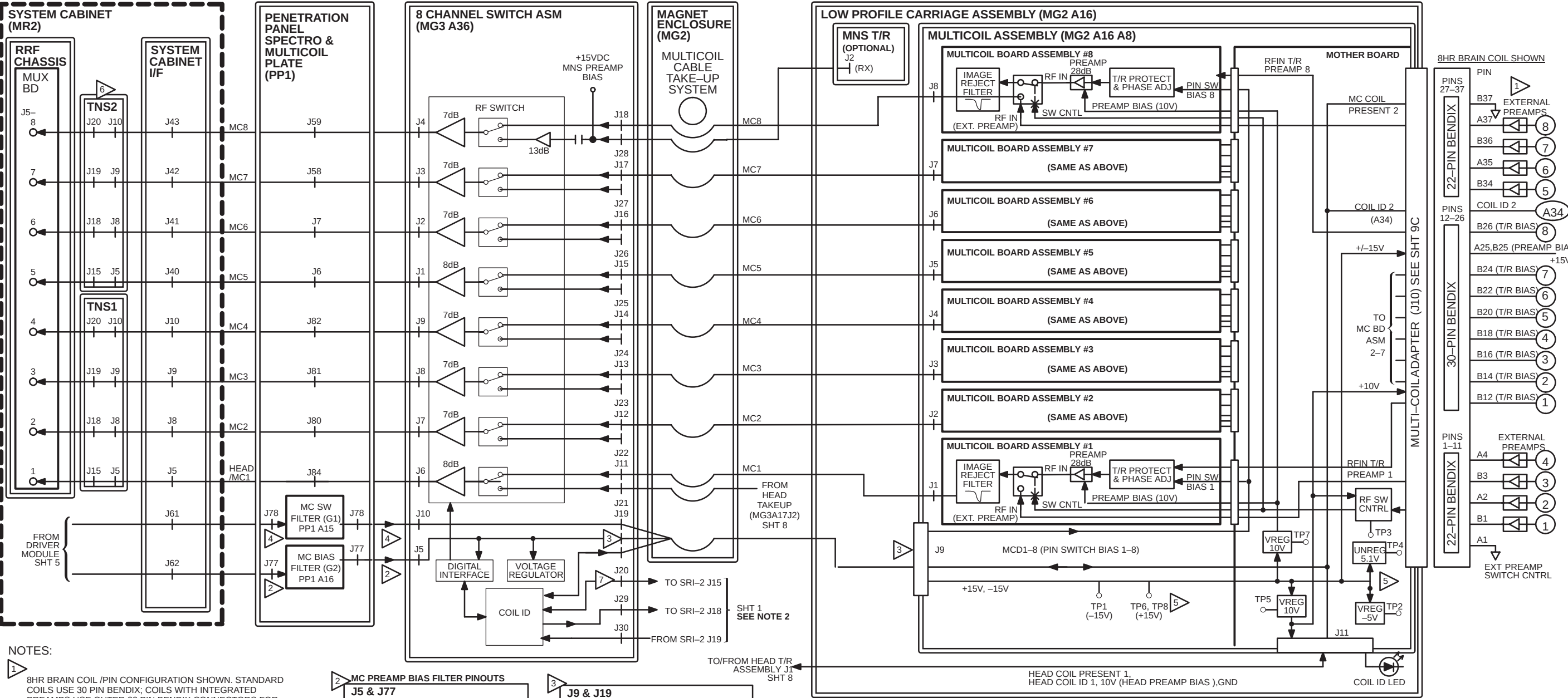


MGD/RRF/RF SUBSYSTEM
SRFD II CABINET – SRFD II RF AMPLIFIER
SHEET 7B OF 10

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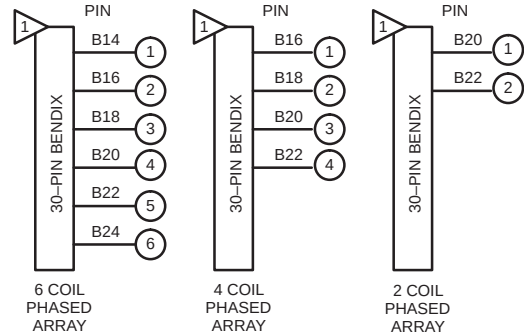
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NOTES:

1 8HR BRAIN COIL /PIN CONFIGURATION SHOWN. STANDARD COILS USE 30 PIN BENDIX; COILS WITH INTEGRATED PREAMPS USE OUTER 22 PIN BENDIX CONNECTORS FOR RF INPUTS AND 30 PIN CONNECTOR FOR COIL DC BIAS.



MGD/RRF/RF SUBSYSTEM – LINUX
MULTI-COIL MODULES

SHEET 9A OF 10

J5 & J77 PIN SIGNAL NAME	
1	XPT_DATA_POS
2	XPT_STROBE_POS
3	XPT_OE_N_POS
4	AGND
5	AGND
6	+15V (NOTE 1)
7	+15V (NOTE 1)
8	+15V (NOTE 1)
9	XPT_DATA_NEG
10	XPT_STROBE_NEG
11	XPT_OE_N_NEG
12	AGND
13	AGND
14	-15V (MC COILS)
15	-15V (MC COILS)

NOTE 1: ANALOG POWER FOR PREAMPS (MC & HEAD), CROSSPOINT & MC COILS.

J9 & J19 PIN(S) SIGNALS	
1,14	-15V
2,15	+15V
3, 4, 13, 23–25	GND
5	MCD8
6	MCD4
7	MCD7
8	MCD3
9	MCD6
10	MCD2
11	MCD5
12	MCD1
16	RED_LED
17	YEL_LED
18	GRN_LED
19	COIL PRESENT #1
20	COIL PRESENT #2
21	COIL ID #1
22	COIL ID #2

J10 & J78 PIN(S) SIGNAL	
1–8	MCD1–8 (NOTE 1)
9	MC CABLE PRESENT_O IN (NOTE 2)
10,11	NO CONNECT
12–19	MCD9–16 (NOTE 3)
20–27	GND
29	MC CABLE PRESENT_1 IN (NOTE 2)
30–37	GND

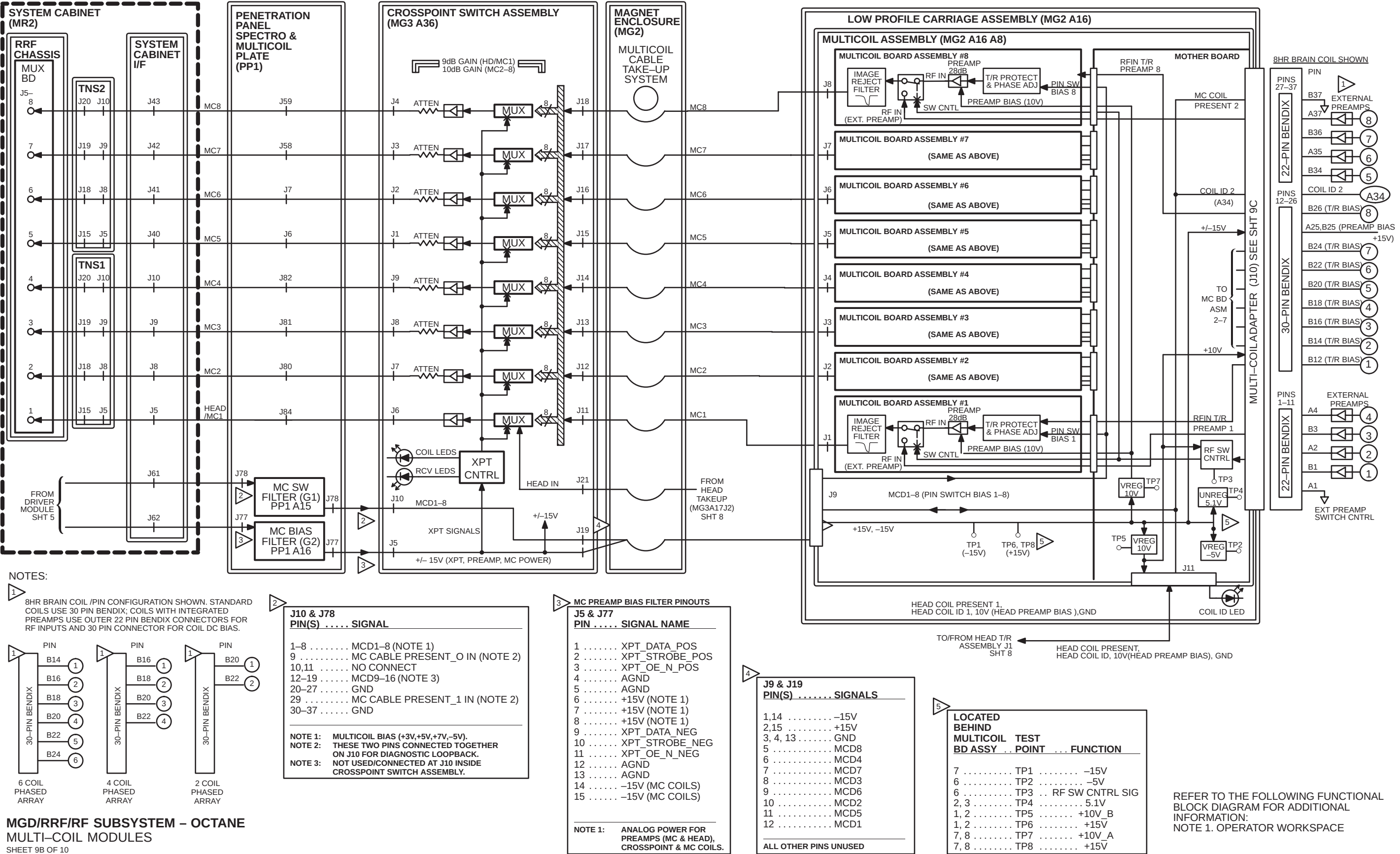
NOTE 1: MULTICOIL BIAS (+3V,+5V,+7V,-5V).
NOTE 2: THESE TWO PINS CONNECTED TOGETHER ON J10 FOR DIAGNOSTIC LOOPBACK.
NOTE 3: NOT USED/CONNECTED AT J10 INSIDE CROSSPOINT SWITCH ASSEMBLY.

LOCATED BEHIND MULTICOIL TEST BD ASSY			POINT	FUNCTION
7	TP1	-15V		
6	TP2	-5V		
6	TP3	RF SW CNTRL SIG		
2, 3	TP4	5.1V		
1, 2	TP5	+10V_B		
1, 2	TP6	+15V		
7, 8	TP7	+10V_A		
7, 8	TP8	+15V		

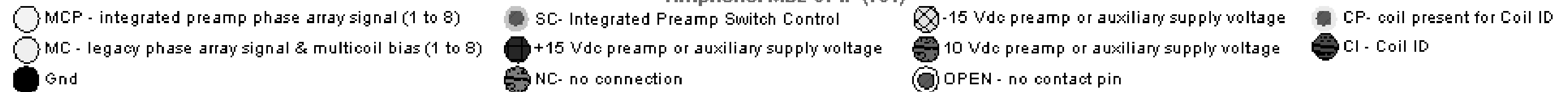
J20 PIN(S) SIGNALS	
1	RED_LED
2	YEL_LED
3	GRN_LED
4	COIL PRESENT #1
5	COIL PRESENT #2
6	24VDC_1
7	24VDC_2
8,9	NO CONNECT

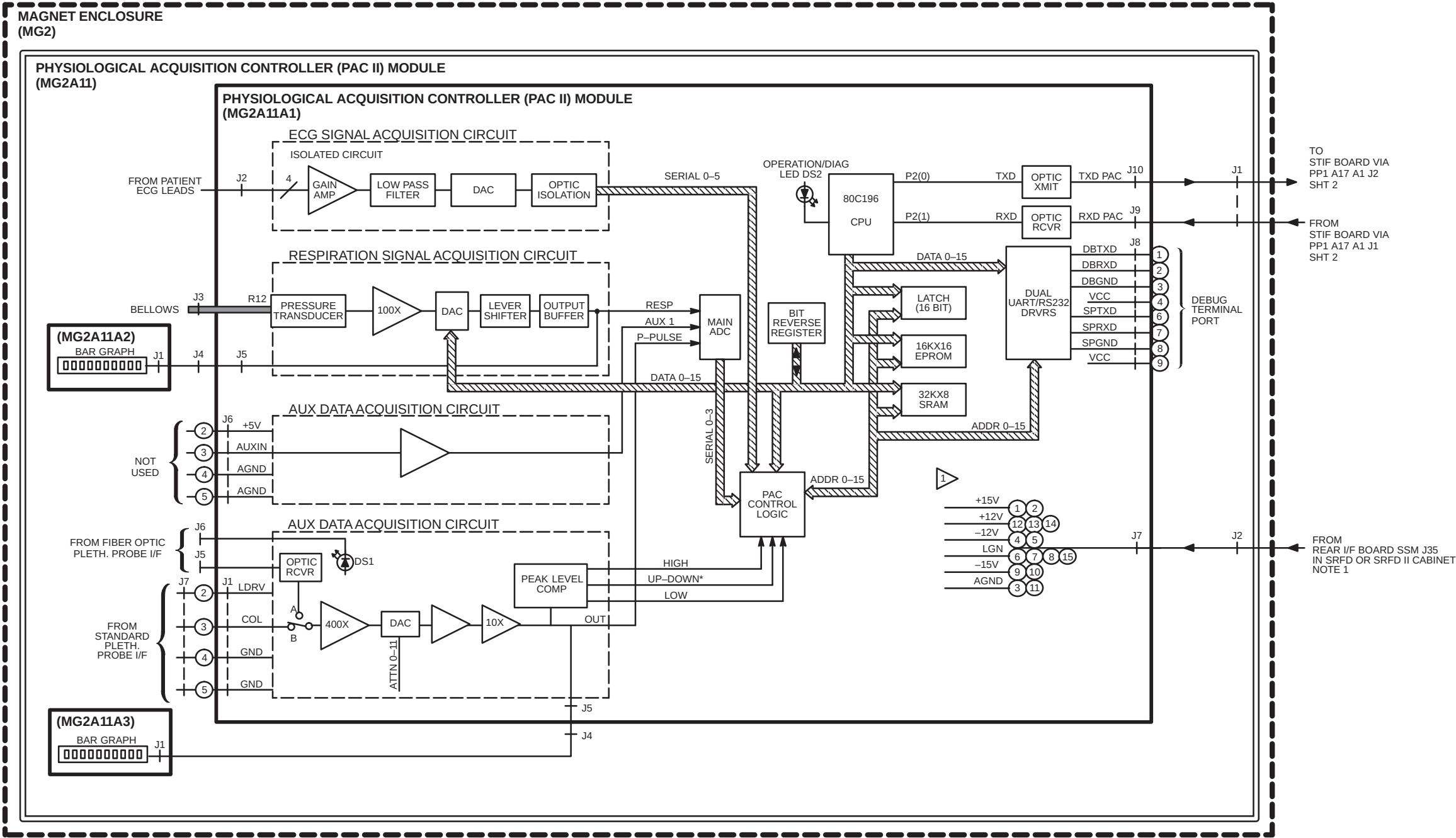
REFER TO THE FOLLOWING FUNCTIONAL BLOCK DIAGRAM FOR ADDITIONAL INFORMATION:

NOTE:
1) OPERATOR WORKSPACE
2) PATIENT HANDLING (LINUX)



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NOTES:

1 → MULTIPLE WIRES/PINS WITH SAME SIGNAL ARE NOT SHOWN

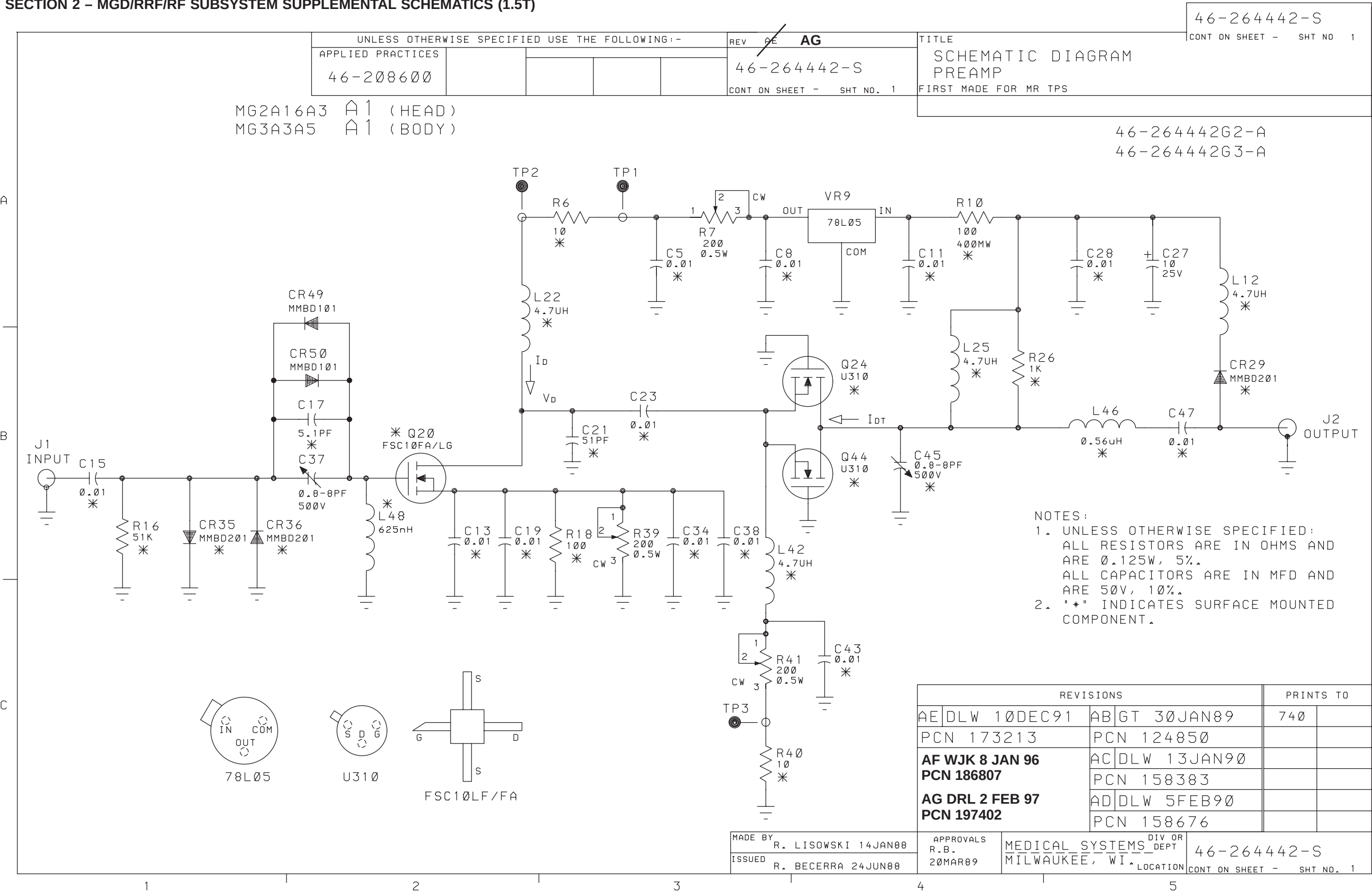
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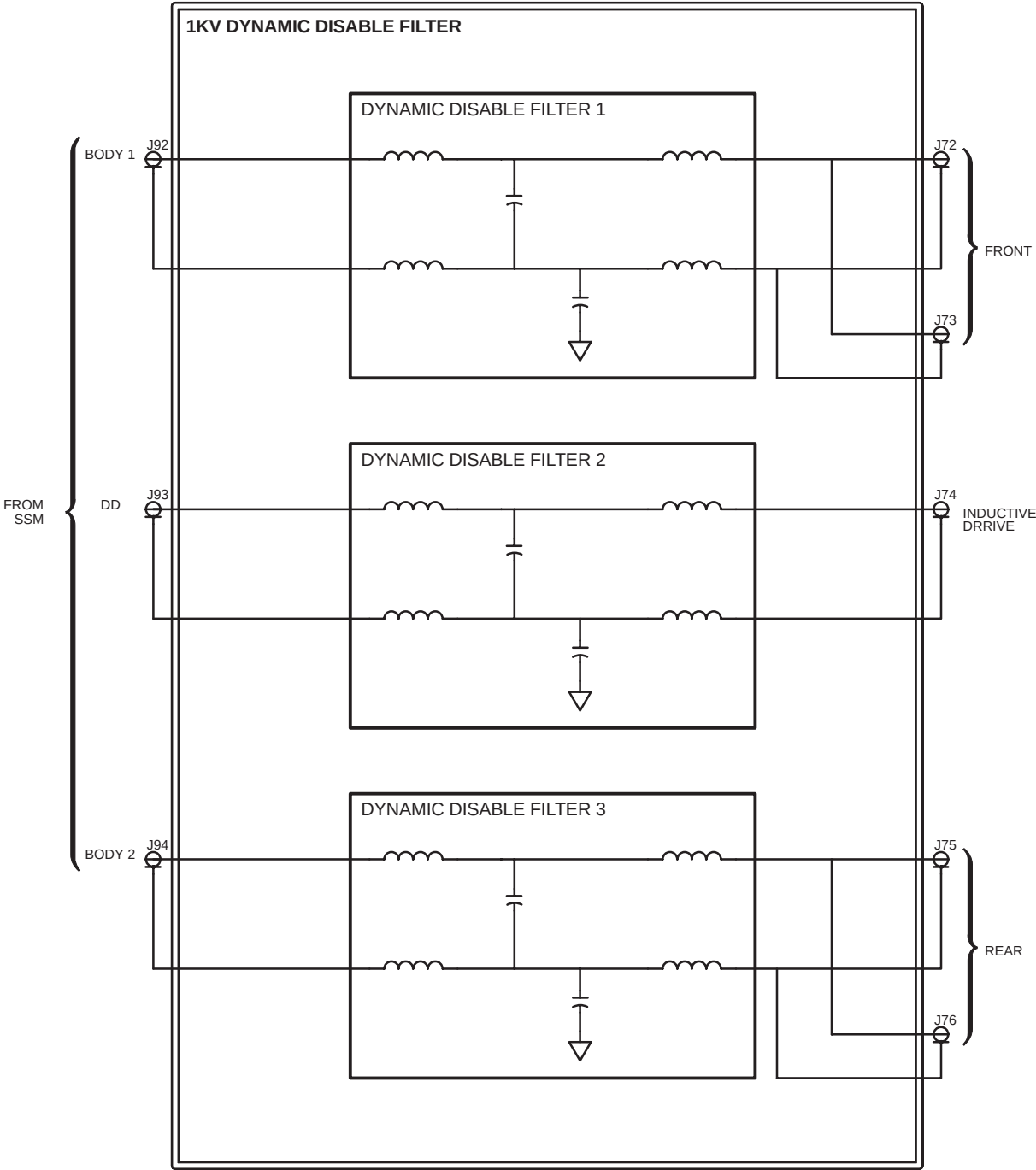
NOTE:

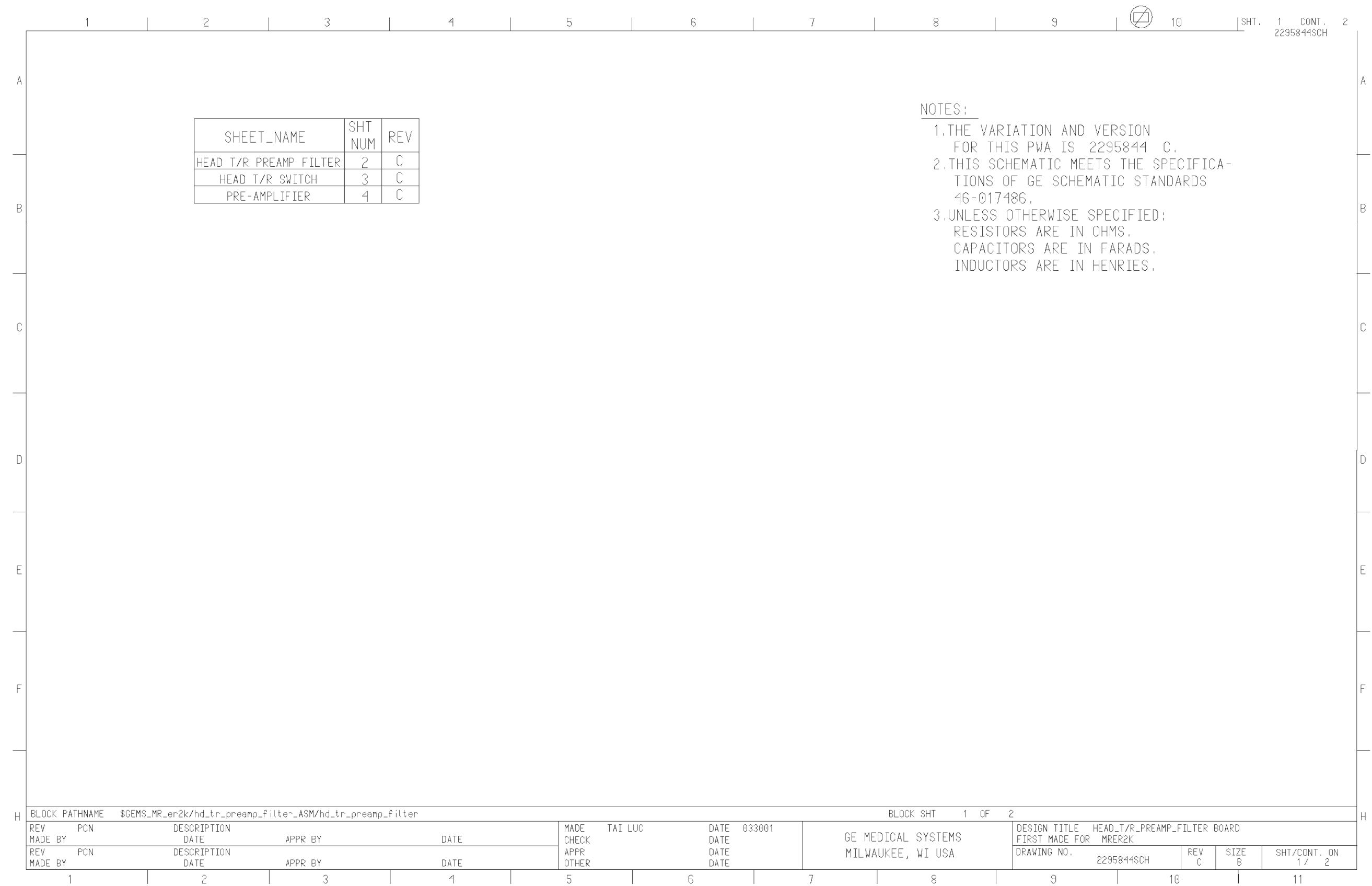
1) PATIENT HANDLING

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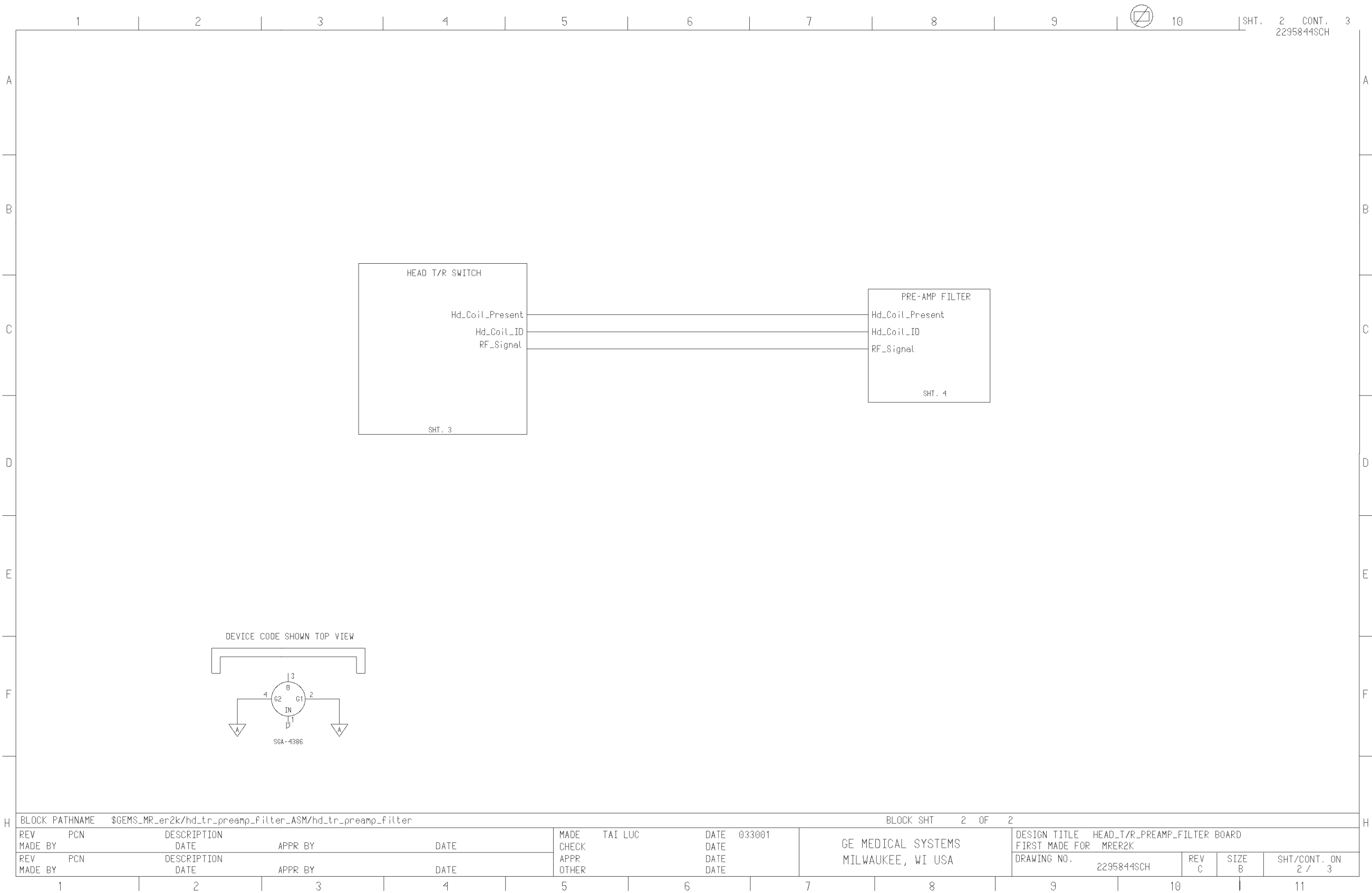
SECTION 2 - MGD/RRF/RF SUBSYSTEM SUPPLEMENTAL SCHEMATICS (1.5T)



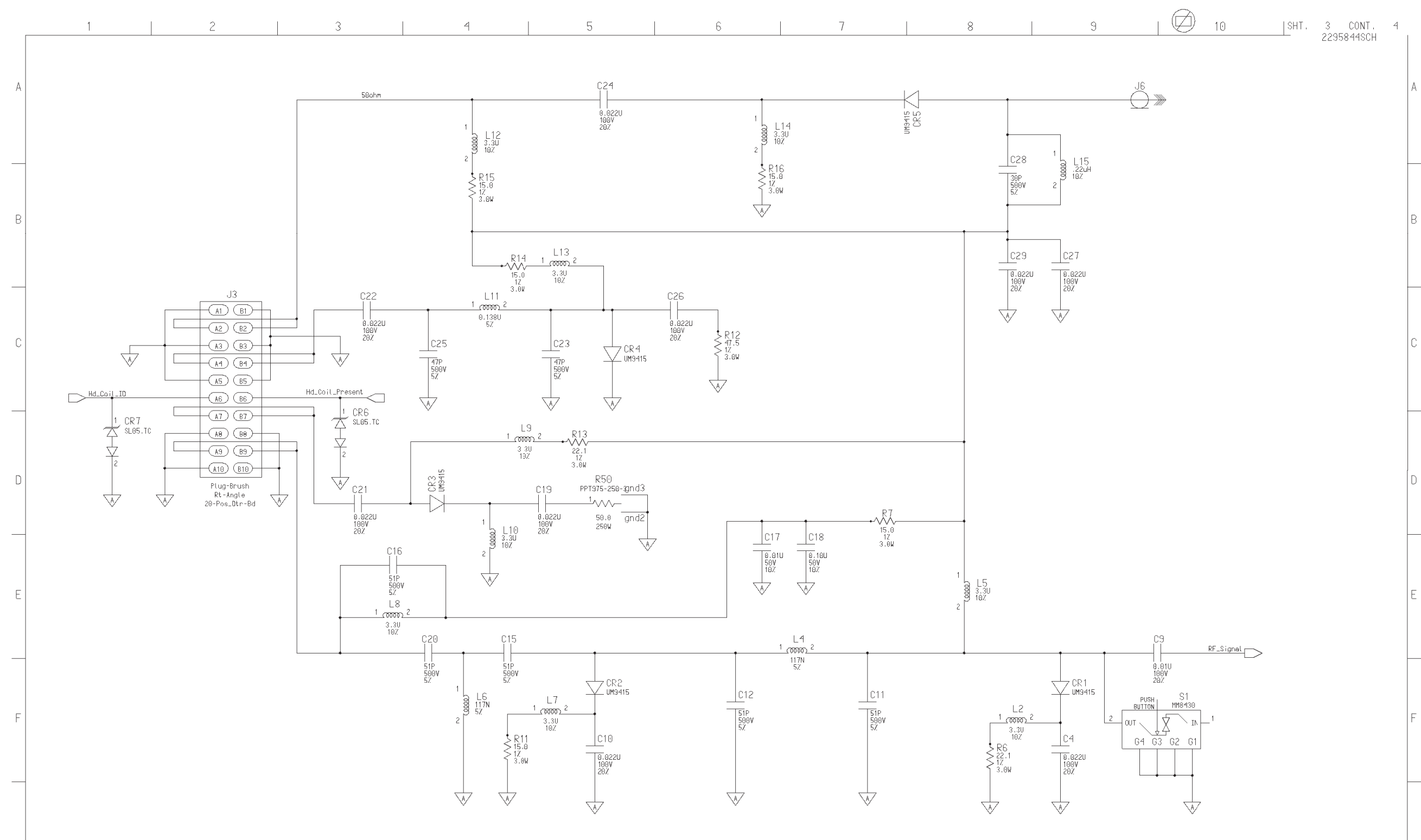




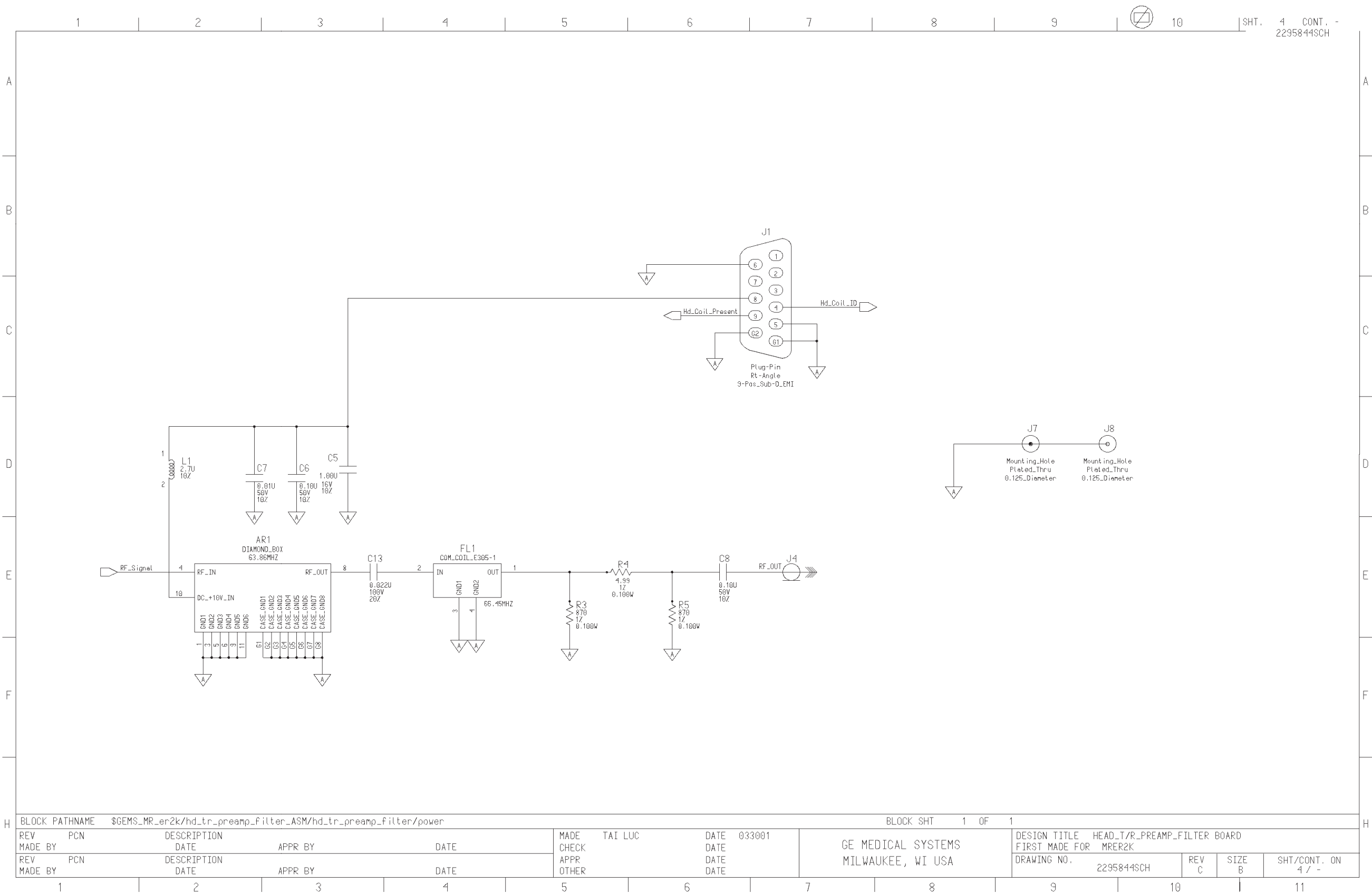
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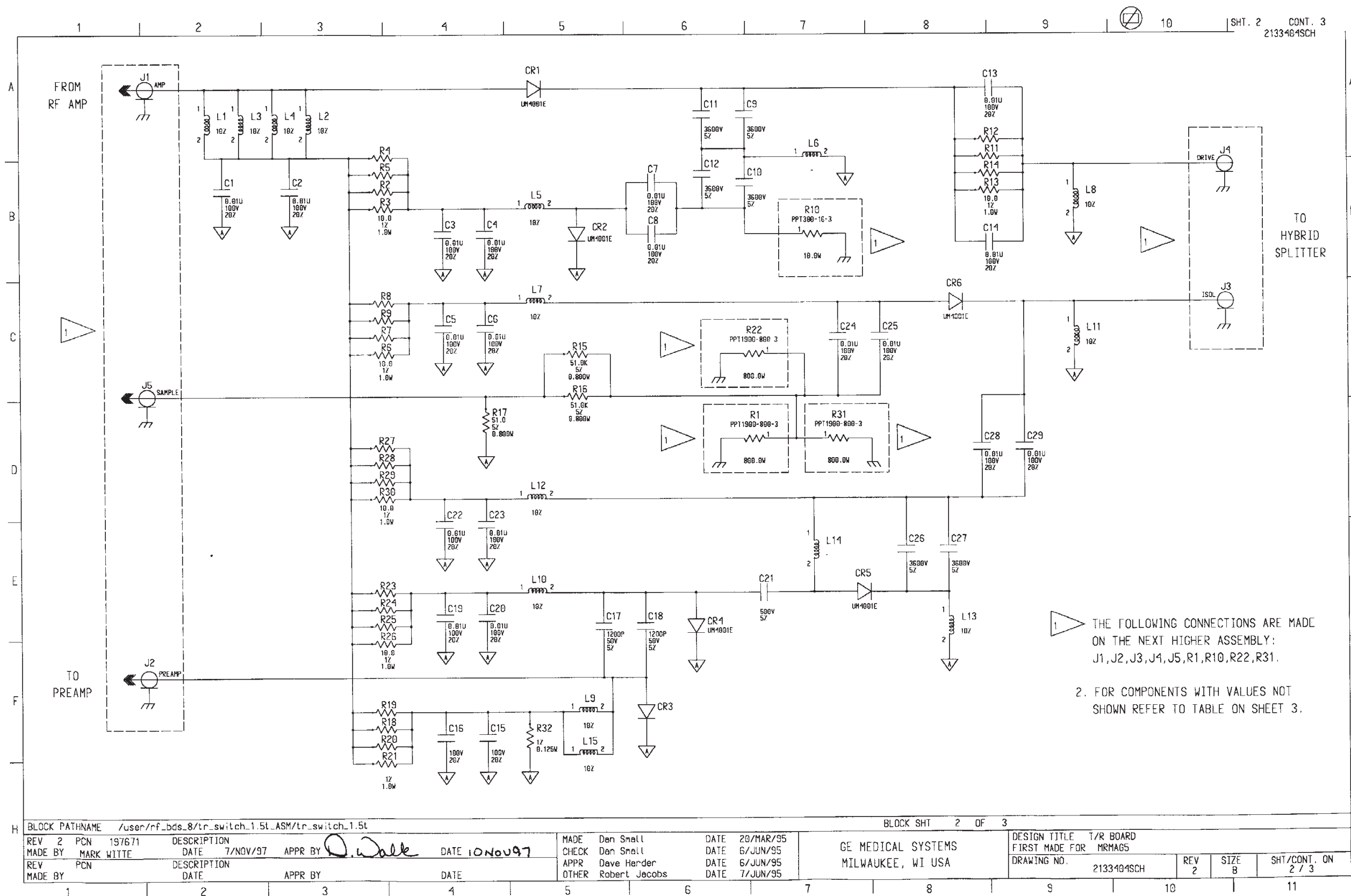
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BLOCK PATHNAME					BLOCK SHT 1 OF 1				
REV	PCN	DESCRIPTION		MADE	DATE 033001		GE MEDICAL SYSTEMS MILWAUKEE, WI USA		
MADE BY		DATE	APPR BY	CHECK	DATE	DATE			
REV	PCN	DESCRIPTION		APPR	DATE		DESIGN TITLE HEAD_T/R_PREAMP_FILTER BOARD FIRST MADE FOR MRER2K		
MADE BY		DATE	APPR BY	OTHER	DATE	DATE			
1		2	3	4	5	6	7	8	9
									10
									11



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COOLING SYSTEM

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For all Chillers and Cooling System variations (SEE Vendor Manuals on MR Service Methods CDROM)

